Chapter 10

Graphs

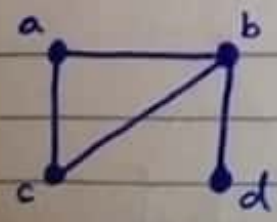
Exercise 10.1

Q: determine whether the graph has directed or undirected edges, whether it has multiple edges and whether it has one ore more loops. Use your answers to determine the type of graph in Table 1 this graph is:

Table 1:

Type	Edges	Multi edges Allowed?	Loops Allowed?
Simple graph	Undirected	No	No
Multigraph	"	Yes	No
Pseudograph	"	Yes	Yes
Simple directed graph	Directed	No	No
Directed multigraph	"	Yes	Yes
Mixed graph	Undirected+directed	Yes	Yes

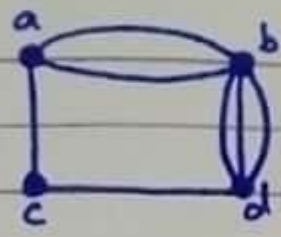
3:



This is a simple graph, has no loop and no multiedges. and also undirected

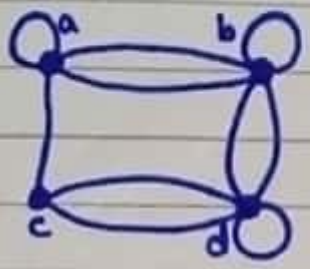
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4:



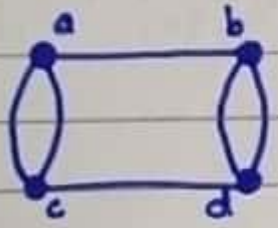
This is multigraph as it is undirected, no loops but has multi edges.

5:



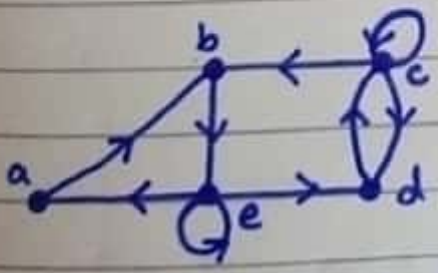
This is pseudograph as it is undirected, has loops and has multi edges.

6:



This is multigraph as it is undirected no loops but has multi edges.

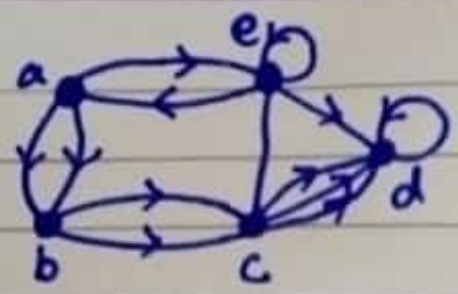
7:



This is directed multigraph as it is directed, has loops and has multiedges.

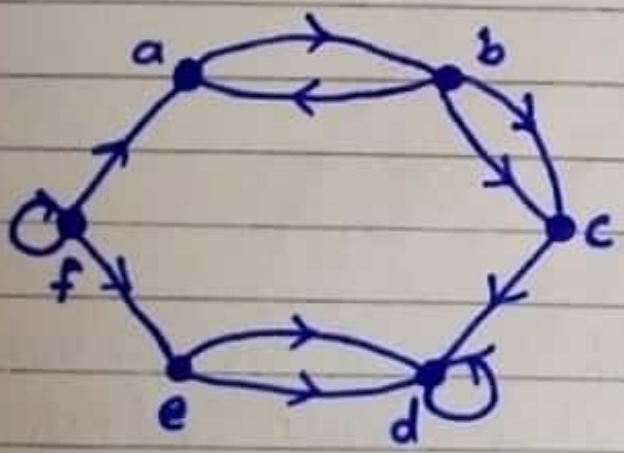
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8:



This is directed multigraph as it is directed, has loops and has multi edges.

9:



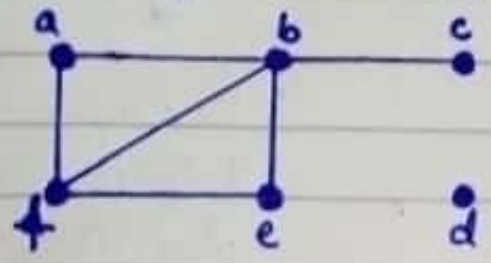
This is directed multigraph as it is directed, has loops and has multi edges.

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Exercise 10.2

Q: Find no of vertices, number of edges, degree of each vertex in given graph. Identify all isolated and pendant vertices.

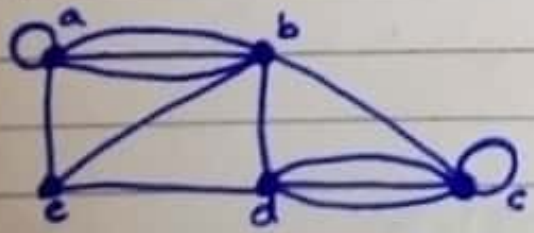
1:



Vertices = 6
Edges = 6
deg(a) = 2
deg(b) = 4
deg(c) = 1
deg(d) = 0
deg(e) = 2
deg(f) = 3
Pendant = c
Isolated = d

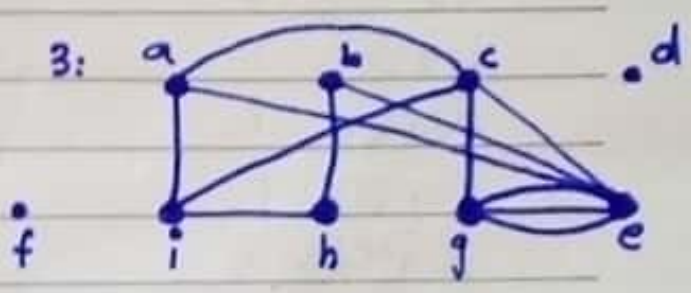
Note:
Vertex of degree zero is isolated and vertex of degree one is pendant.

2:



Vertices = 5 , Edges = 13
deg(a) = 6 , deg(b) = 6
deg(c) = 6 , deg(d) = 5
deg(e) = 3
Pendant = No
Isolated = No

3:

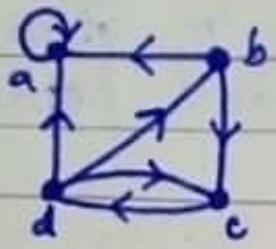


Vertices = 9
Edges = 12
Deg(a) = 3
Deg(b) = 2
Deg(c) = 4
Deg(d) = 0
Deg(e) = 6
Deg(f) = 0
Deg(g) = 4
Deg(h) = 2
Deg(i) = 3
Pendant = No
Isolated = f, d

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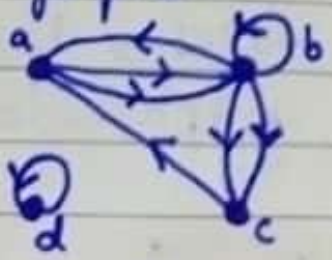
Q: Determine number of vertices, edges, find in-degree and out degree of given graphs.

7:



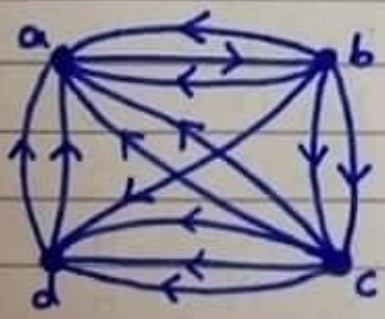
Vertices = 4
Edges = 7
$deg^-(a) = 3$
$deg^+(a) = 1$
$deg^-(b) = 1$
$deg^+(b) = 2$
$deg^-(c) = 2$
$deg^+(c) = 1$
$deg^-(d) = 1$
$deg^+(d) = 3$

8:



Vertices = 4
Edges = 8
$deg^-(a) = 2$
$deg^+(a) = 2$
$deg^-(b) = 3$
$deg^+(b) = 4$
$deg^-(c) = 2$
$deg^+(c) = 1$
$deg^-(d) = 1$
$deg^+(d) = 1$

9:



Vertices = 5
Edges = 13
$deg^-(a) = 6$
$deg^+(a) = 1$
$deg^-(b) = 1$
$deg^+(b) = 5$
$deg^-(c) = 2$
$deg^+(c) = 5$
$deg^-(d) = 4$
$deg^+(d) = 2$

e:

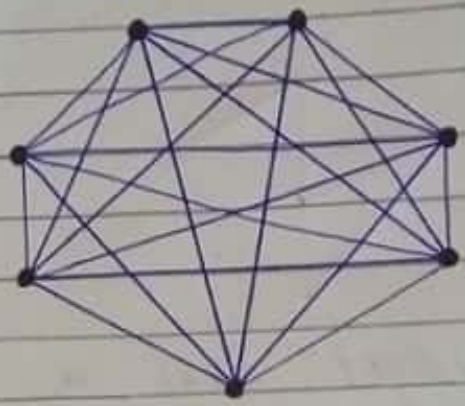
$deg^-(e) = 0$
$deg^+(e) = 0$

Note:
(−) show in degree
(+) show out degree

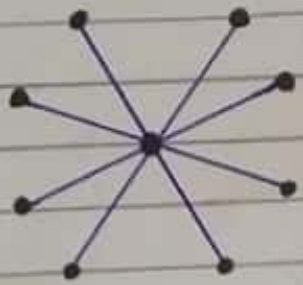
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20: Draw these graphs.

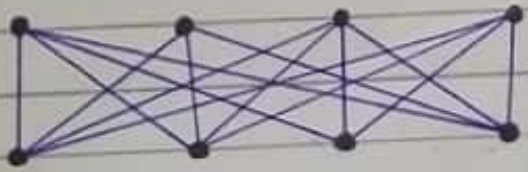
a) K_7



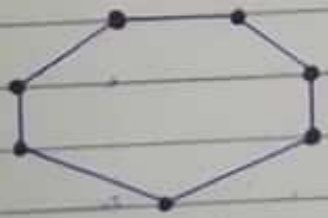
b) $K_{1,8}$



c) $K_{4,4}$



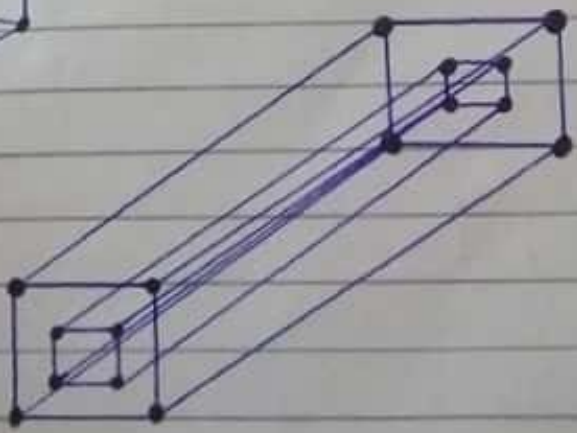
d) C_7



e) W_7

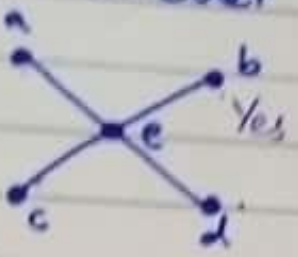


f) Q_4

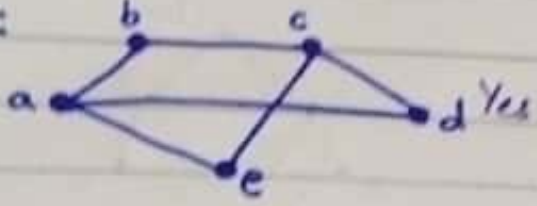


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Determine whether graph is bipartite.

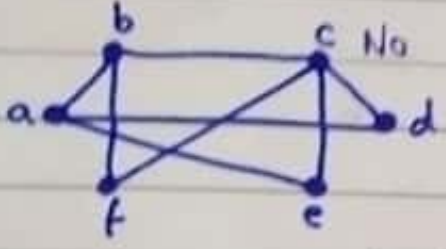
21: 

Yes

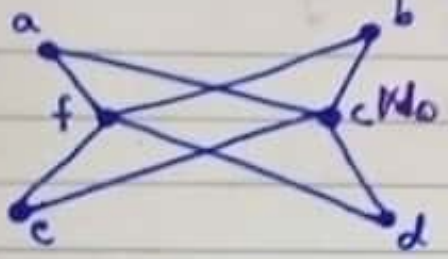
22: 

Yes

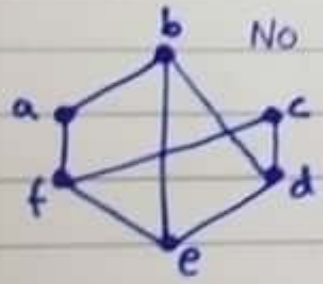
Vertices are odd then it is bipartite and if vertices are even then it is not bipartite.

23: 

No

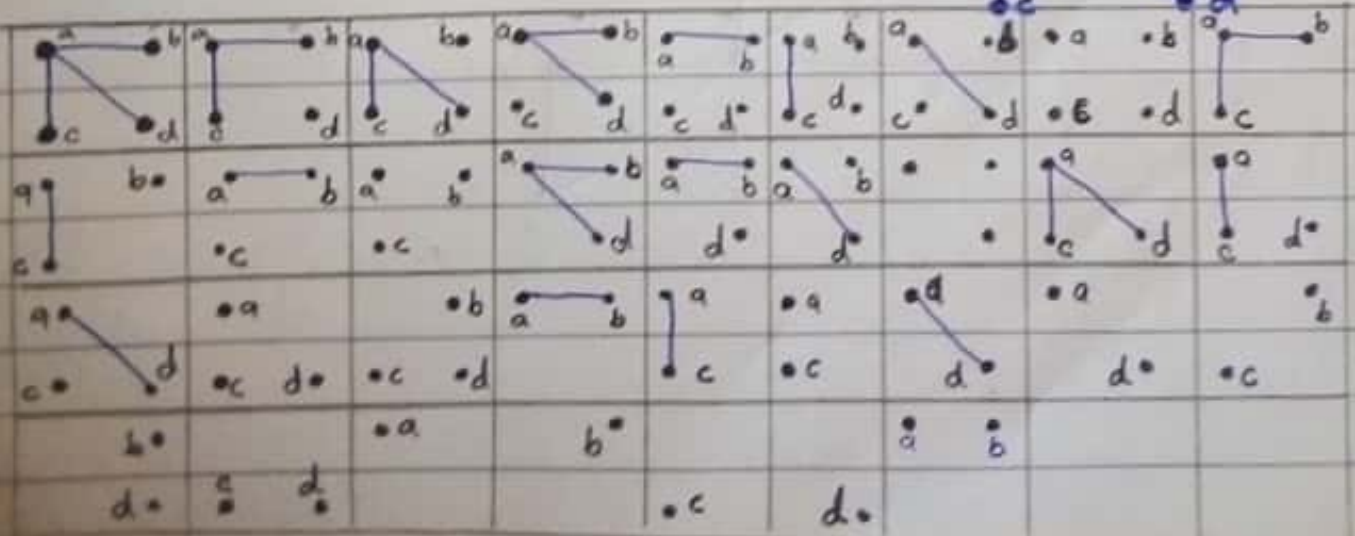
*24: 

No

25: 

No

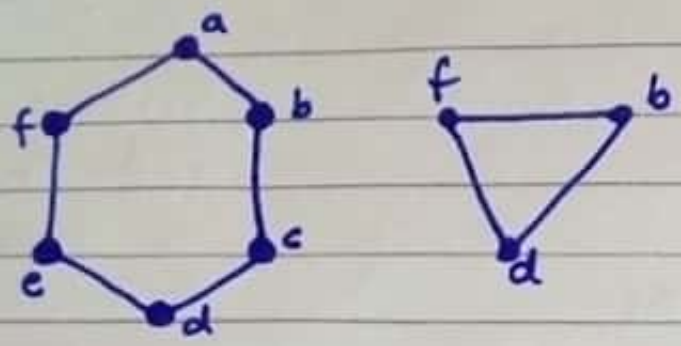
53: Draw all subgraph of this 

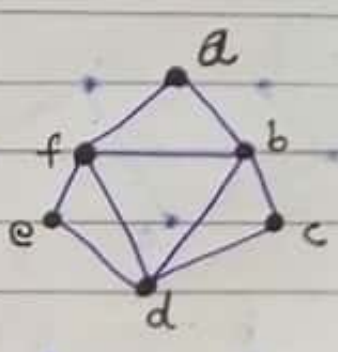


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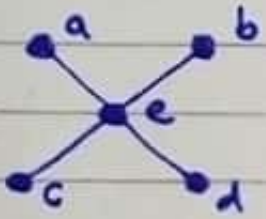
Find union of given pair of graph.

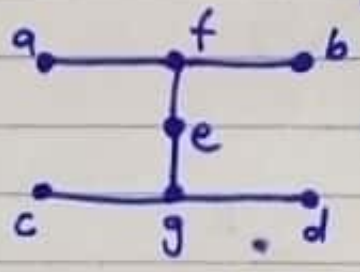
58:

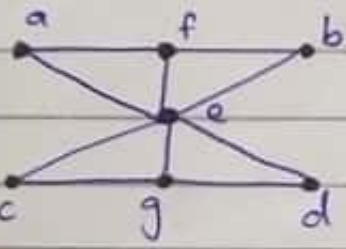




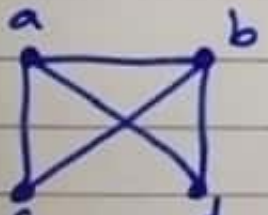
59:

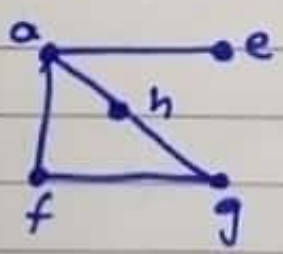


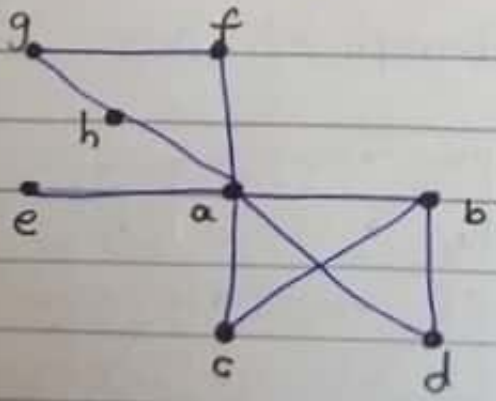




60:

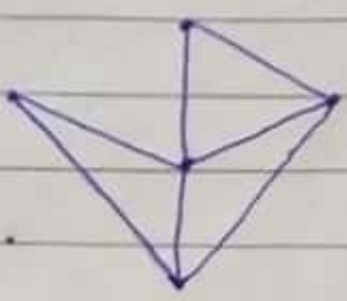




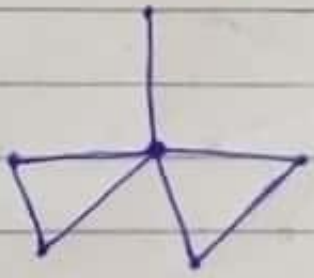


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42: How many edges does a graph have if its degree sequence is 4, 3, 3, 2, 2? Draw such graph



43: How many edges does a graph have if its degree sequence is 5, 2, 2, 2, 2, 1

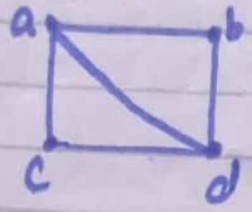


44:

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Exercise # 10.3

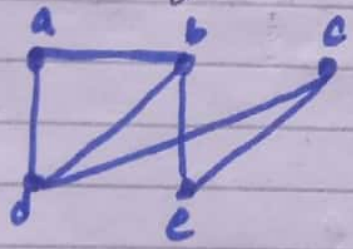
In Exercises 1-4 use an adjancy list to represent the given graph.

1. 

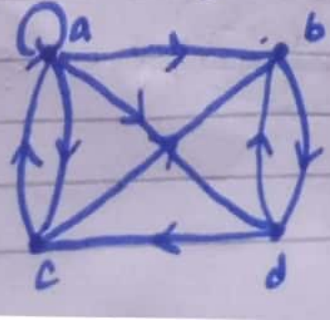
Adjancy lists are lists for lists. The adjancy list for an undirected graph is simply the list of vertices of the given graph, together with the list of vertices adjacent to it.

The list for this graph is follows.

vertex	Adjacent vertex
a	b, c, d
b	a, d
c	a, d
d	a, b, c

2. 

vertex	Adjacent vertex
a	b, d
b	a, d, e
c	d, e
d	a, b, c
e	b, c

3. 

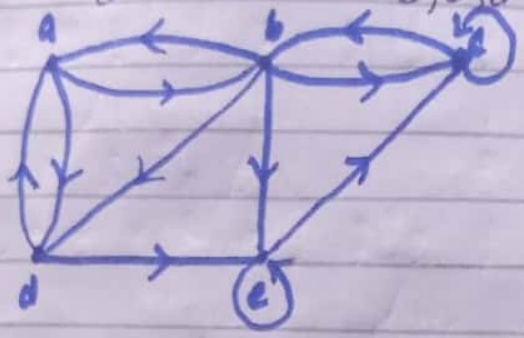
In directed

Check out degree

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vertex	Adjacent vertex
a	a, b, c, d
b	d
c	a, b
d	b, a, d

4.



vertex	Adjacent vertex
a	b, d
b	a, c, d, e
c	b, c
d	a, e
e	c, e

5. Represent the graph in Ex 1 with an adjancy matrix.

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

6. Represent the graph in Ex 2 with an adj— matrix:-

$$\begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

7. Represent the graph in Ex 3 with an adjancy matrix.

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$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

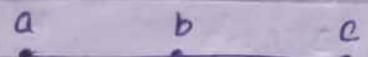
8. Represent the graph in Ex 4 with an adjancy matrix.

$$\begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix}$$

In Exercises 10-12 draw graph with the given adjancy matrix.

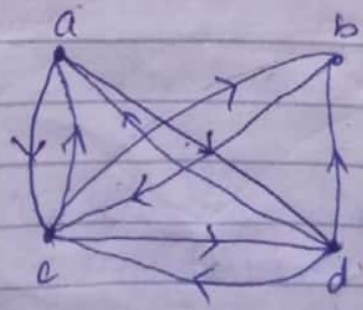
10. $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

This graph is undirected becs $A^T = A$ so it is symmetric.



11. $\begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{array}$

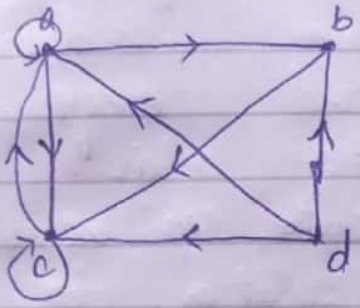
This graph is directed because in this $A^T \neq A$ so it is anti-symmetric.



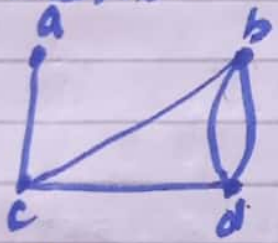
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12. $\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}$

This is also directed.

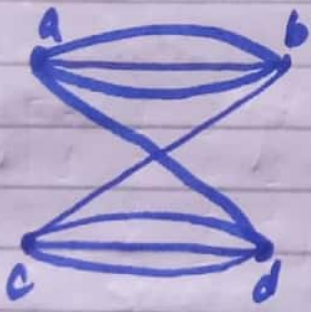


In Exercises 13-15 represent the given graph using an adjacency matrix.

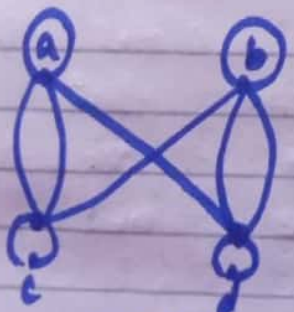
13. 

$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 2 \\ 1 & 1 & 0 & 1 \\ 0 & 2 & 1 & 0 \end{bmatrix}$

No of adjacent vertices =

14. 

$\begin{bmatrix} 0 & 3 & 0 & 1 \\ 3 & 0 & 1 & 0 \\ 0 & 1 & 0 & 3 \\ 1 & 0 & 3 & 0 \end{bmatrix}$

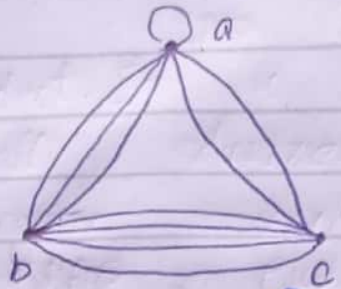
15. 

$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 2 \\ 2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$

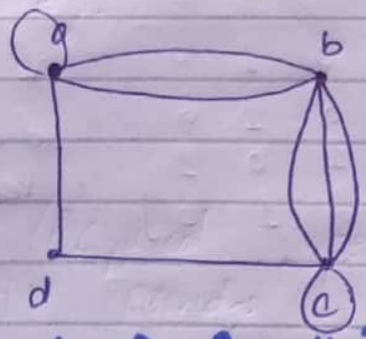
In exercises 16-18 draw an undirected graph represented by given adjacency matrix.

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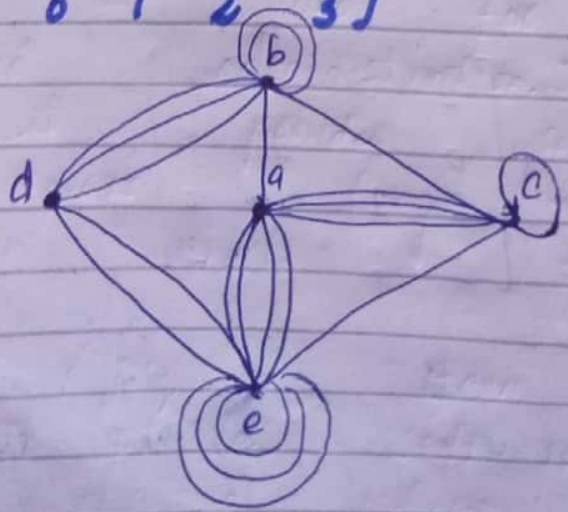
16. $\begin{bmatrix} 1 & 3 & 2 \\ 3 & 0 & 4 \\ 2 & 4 & 0 \end{bmatrix}$



17. $\begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 0 & 3 & 0 \\ 0 & 3 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$



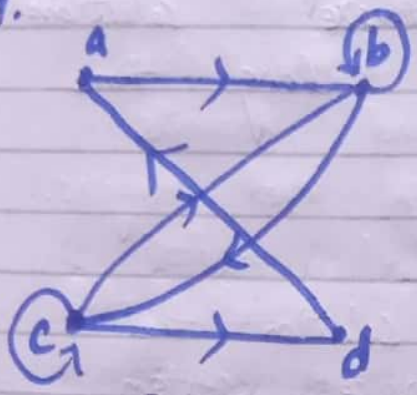
18. $\begin{bmatrix} 0 & 1 & 3 & 0 & 4 \\ 1 & 2 & 1 & 3 & 0 \\ 3 & 1 & 1 & 0 & 1 \\ 0 & 3 & 0 & 0 & 2 \\ 4 & 0 & 1 & 2 & 3 \end{bmatrix}$



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In Exercises 19-21 find adjancy matrix in alphabetical order.

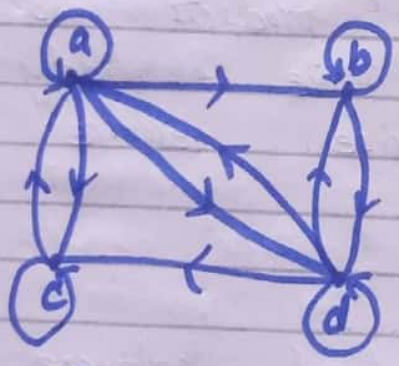
19.



loop is counted as 1 in matrix.

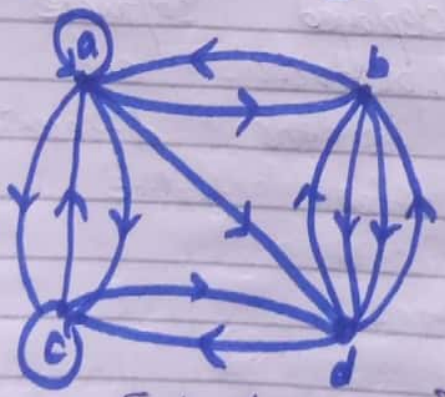
$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

20.



$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

21.



If there are two edges from a to c then it 2 and loop is 1.

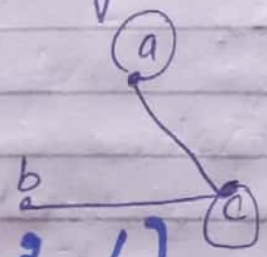
$$\begin{bmatrix} 1 & 1 & 2 & 1 \\ 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & 1 \\ 0 & 2 & 1 & 0 \end{bmatrix}$$

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In Exercise 22-24 draw graph represented by given adjacency matrix.

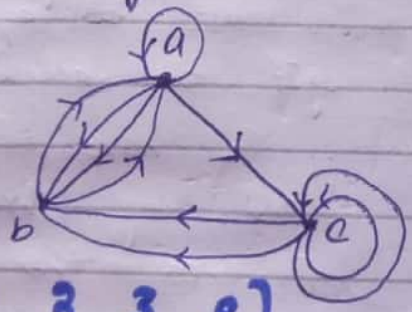
22. $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

This graph is undirected bcs $A^T = A$



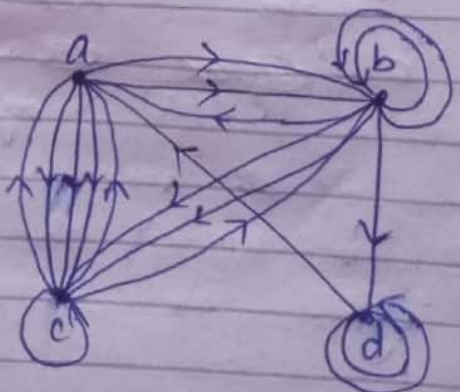
23. $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 0 & 0 \\ 0 & 2 & 2 \end{bmatrix}$

This graph is directed bcs $A^T \neq A$



24. $\begin{bmatrix} 0 & 2 & 3 & 0 \\ 1 & 2 & 2 & 1 \\ 2 & 1 & 1 & 0 \\ 1 & 0 & 0 & 2 \end{bmatrix}$

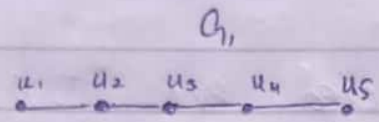
This graph is directed bcs $A^T \neq A$

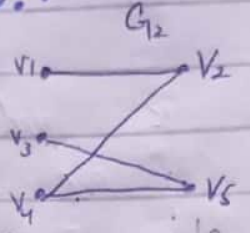


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In Exercises 34-44 determine whether the given pair of graphs is isomorphic. Exhibit an isomorphism or provide a rigorous argument that none exist.

34.





Solution:-

1. No of vertices in both graphs is same
2. No of edges in both graphs is same
3. $\deg(u_1) = 1$ $\deg(v_1) = 1$
   $\deg(u_2) = 2$ $\deg(v_2) = 2$
   $\deg(u_3) = 2$ $\deg(v_3) = 1$
   $\deg(u_4) = 2$ $\deg(v_4) = 2$
   $\deg(u_5) = 1$ $\deg(v_5) = 2$
   8 8
4. In G_1 there are two odd and three even degrees. In G_2 there are also two odd and three even degrees so condition 3 is satisfied.
5. $f(u_1) = v_1$
   $f(u_2) = v_2$
   $f(u_3) = v_4$
   $f(u_4) = v_5$ and
   $f(u_5) = v_3$

Conditions of isomorphism

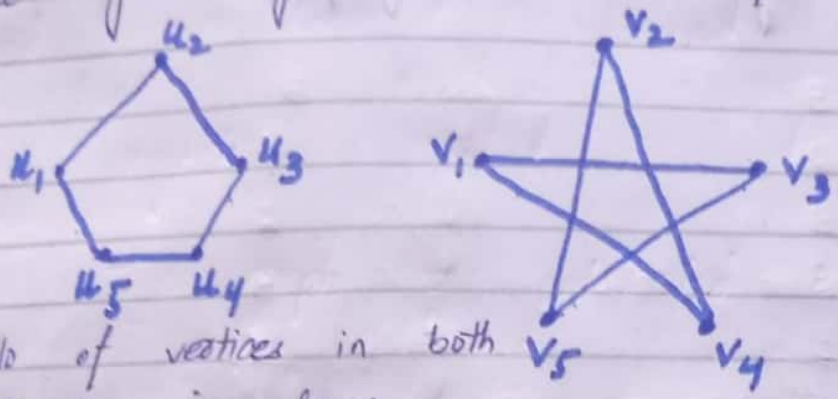
1. No of vertices must be same in both
2. No of edges must be same in both
3. Sum of degrees must be equal
4. eg If two vertices in G_1 have odd degree and any three have even then there must be two vertices in G_2 whose degree is odd and three vertices whose degree is even.
5. $f(G_1) = G_2$ degree seq.

If all above conditions are satisfied then two graphs are Isomorphic.

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Since all the conditions are satisfied so the given graphs are isomorphic.

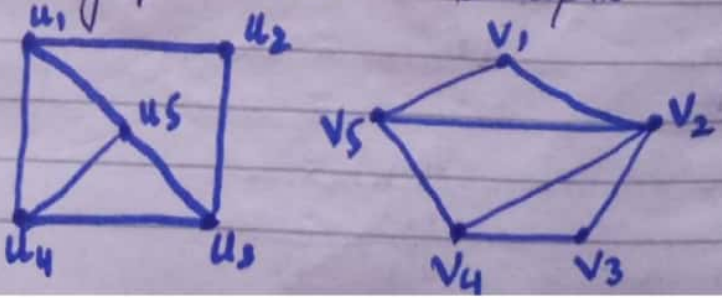
35.



1. No of vertices in both graphs is same.
2. No of edges in both graphs is same.
3. $\deg(u_1) = 2$ $\quad$ $\deg(v_1) = 2$
   $\deg(u_2) = 2$ $\quad$ $\deg(v_2) = 2$
   $\deg(u_3) = 2$ $\quad$ $\deg(v_3) = 2$
   $\deg(u_4) = 2$ $\quad$ $\deg(v_4) = 2$
   $\deg(u_5) = 2$ $\quad$ $\deg(v_5) = 2$
   10 $\quad$ 10
4. All of the five vertices in both the graphs have same number of even degrees.
   In G_1 if there are 5 even degrees the in G_2 it is also.
5. $f(u_1) = v_1$
   $f(u_2) = v_3$
   $f(u_3) = v_5$
   $f(u_4) = v_2$
   $f(u_5) = v_4$

Since all the conditions are satisfied so the given graphs are isomorphic.

36.



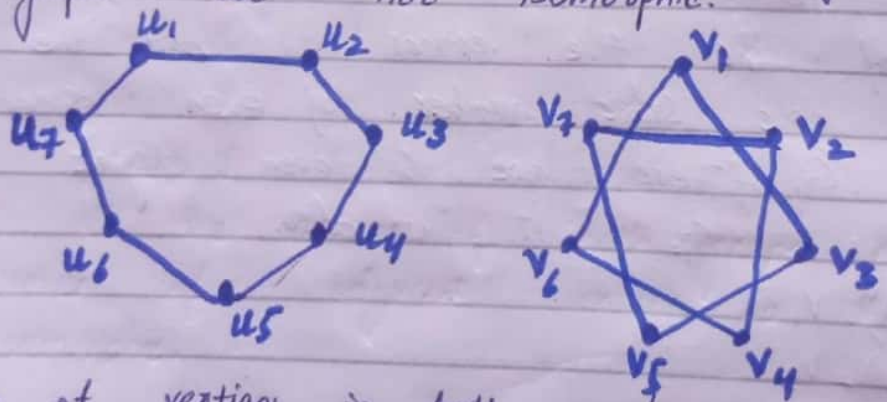
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1. No of vertices in both graphs is same.
2. No of edges in both graphs is same.
3.

$\deg(u_1) = 3$	$\deg(v_1) = 2$
$\deg(u_2) = 2$	$\deg(v_2) = 4$
$\deg(u_3) = 3$	$\deg(v_3) = 2$
$\deg(u_4) = 3$	$\deg(v_4) = 3$
$\deg(u_5) = \frac{3}{14}$	$\deg(v_5) = \frac{3}{14}$

4. There are 4 odd and 1 even degree in G_1 and there are Two odd and 3 even degrees in G_2. As the numbers of odd and even degrees in both G_1 and G_2 so, the condition 4 is not satisfied So, the given graphs are not isomorphic.

37.



1. No of vertices in both graphs is same.
2. No of edges in both graphs is same.
3.

$\deg(u_1) = 2$	$\deg(v_1) = 2$
$\deg(u_2) = 2$	$\deg(v_2) = 2$
$\deg(u_3) = 2$	$\deg(v_3) = 2$
$\deg(u_4) = 2$	$\deg(v_4) = 2$
$\deg(u_5) = 2$	$\deg(v_5) = 2$
$\deg(u_6) = 2$	$\deg(v_6) = 2$
$\deg(u_7) = \frac{2}{14}$	$\deg(v_7) = \frac{2}{14}$

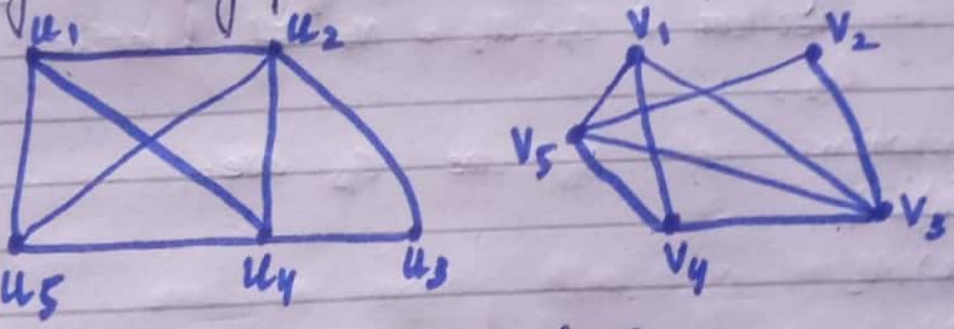
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4. No of even degrees in G_1 is 7 and in G_2 no of even degrees is also 7. There is no odd degree in both. So condition 4 is satisfied.

5. $f(u_1) = v_1$, $f(u_2) = v_3$, $f(u_3) = v_5$, $f(u_4) = v_7$, $f(u_5) = v_2$, $f(u_6) = v_4$, $f(u_7) = v_6$

As all the conditions are satisfied so given graphs are isomorphic.

38.



1. No of vertices are same.
2. No of edges are same.
3. $\deg(u_1) = 3$, $\deg(u_2) = 4$, $\deg(u_3) = 2$, $\deg(u_4) = 4$, $\deg(u_5) = 3$, $\deg(v_1) = 3$, $\deg(v_2) = 2$, $\deg(v_3) = 4$, $\deg(v_4) = 3$, $\deg(v_5) = 4$

16

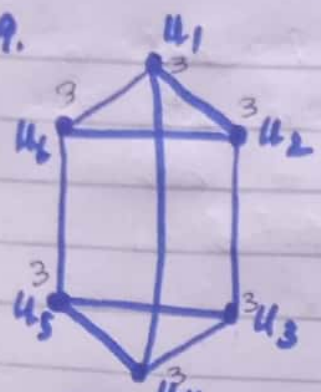
4. No of even degree in G_1 is 3 and odd is 2 and in G_2 even are 3 and odd are 2. So C-4 is satisfied.

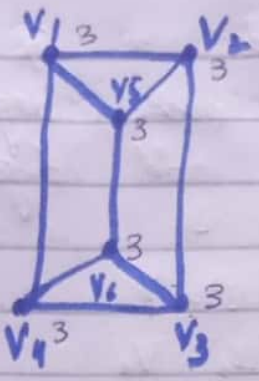
5. $f(u_1) = v_1$, $f(u_2) = v_5$, $f(u_3) = v_2$, $f(u_4) = v_3$, $f(u_5) = v_4$

As all the conditions are satisfied so given graphs are isomorphic.

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39.





Shortly answer is :-

* (This is second method it can also solved by previous one)

Solution:- These two graphs are isomorphic bcs:-

i) No of edges in $G_1 = 9 =$ No of edges in G_2

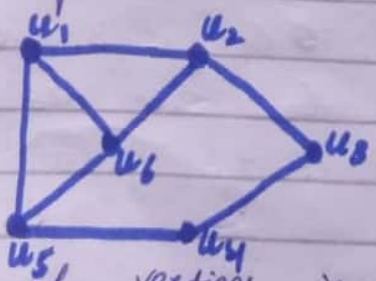
ii) No of vertices in $G_1 = u_6 =$ No of vertices in $G_2 = V_6$

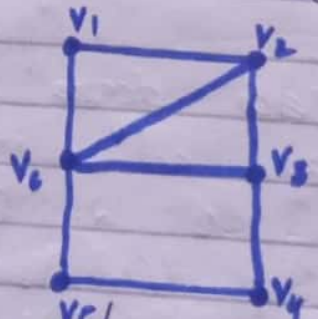
iii) 7 somorphisms of G_1 are G_2 are:-

$f(u_1) = V_1$
$f(u_2) = V_2$
$f(u_3) = V_3$
$f(u_4) = V_4$
$f(u_5) = V_5$
$f(u_6) = V_6$

As all the vertices of both graph have same number of adjacent vertices so they are isomorphic.

40.





1. No of vertices in graph is same.
2. No of edges is also same.
3. $\deg(u_1) = 3$ $\deg(u_4) = 2$
   $\deg(u_2) = 3$ $\deg(u_5) = 3$
   $\deg(u_3) = 2$ $\deg(u_6) = 3$

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$deg(v_1) = 2$ $deg(v_4) = 2$

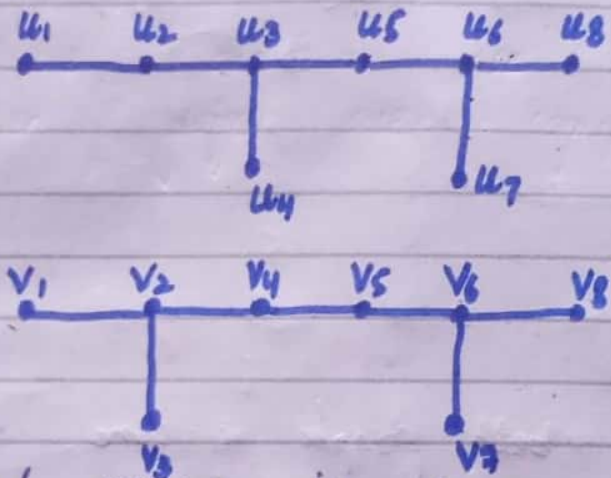
$deg(v_2) = 3$ $deg(v_5) = 2$

$deg(v_3) = 3$ $deg(v_6) = 3$

16 16

4. The number of even and odd degrees in given graphs is not same so they are not ismorphic.

41.



1. No of vertices is same.
2. No of edges is also same.
3. $deg(u_1) = 1$ $deg(v_1) = 1$

$deg(u_2) = 2$ $deg(v_2) = 3$

$deg(u_3) = 3$ $deg(v_3) = 1$

$deg(u_4) = 1$ $deg(v_4) = 2$

$deg(u_5) = 2$ $deg(v_5) = 2$

$deg(u_6) = 3$ $deg(v_6) = 3$

$deg(u_7) = 1$ $deg(v_7) = 1$

$deg(u_8) = 1$ $deg(v_8) = 1$

14 14

4. No of odd and even degrees are same.
5. $f(u_1) =$ Not possible $f(u_5) =$ Not possible

$f(u_2) =$ Not possible $f(u_6) = V_6$

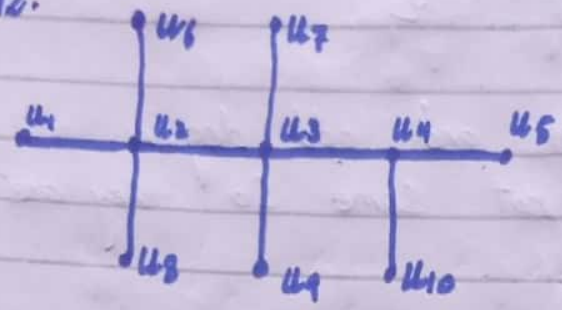
$f(u_3) =$ Not possible $f(u_7) = V_7$

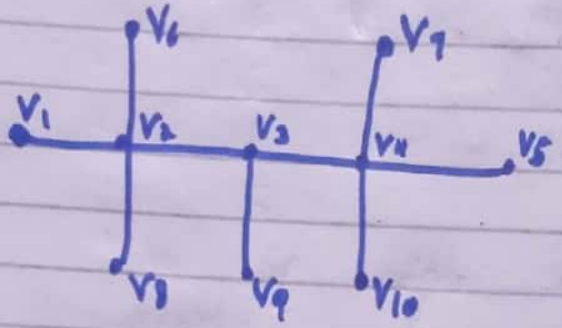
$f(u_4) = V_3$ $f(u_8) = V_8$

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As the condition 5 is not satisfied so G_1 and G_2 are not isomorphism.

42.





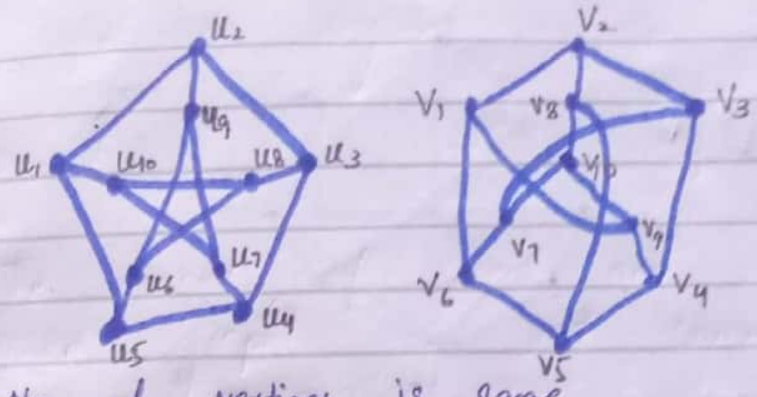
1. No of vertices in both graphs is same.
2. No of edges is also same.
3. $\deg(u_1) = 1$ $\deg(v_1) = 1$
   $\deg(u_2) = 4$ $\deg(v_2) = 4$
   $\deg(u_3) = 4$ $\deg(v_3) = 3$
   $\deg(u_4) = 3$ $\deg(v_4) = 4$
   $\deg(u_5) = 1$ $\deg(v_5) = 1$
   $\deg(u_6) = 1$ $\deg(v_6) = 1$
   $\deg(u_7) = 1$ $\deg(v_7) = 1$
   $\deg(u_8) = 1$ $\deg(v_8) = 1$
   $\deg(u_9) = 1$ $\deg(v_9) = 1$
   $\deg(u_{10}) = 1$ $\deg(v_{10}) = 1$

   $\frac{18}{18}$

4. The no of even and odd degrees is same.
5. $f(u_1) = V_1$
   $f(u_2)$ not possible
   So, they are not Isomorphic.

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43.

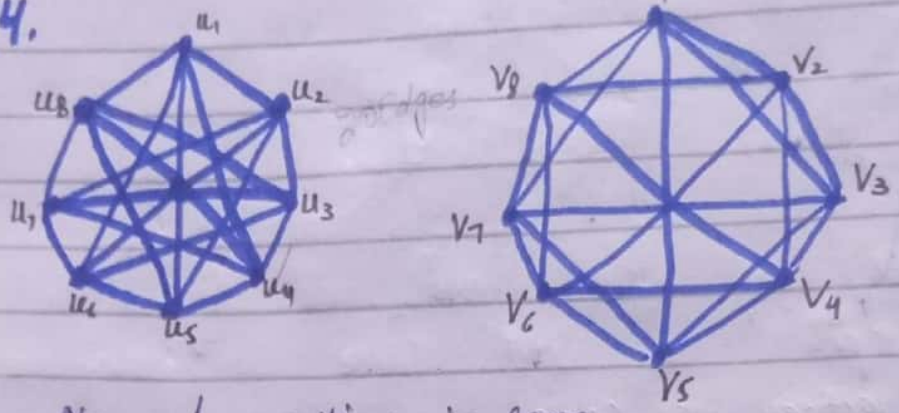


1. No of vertices is same.
2. No of edges is also same.
3. $\deg(u_1) = 3$ $\quad$ $\deg(v_1) = 3$
   $\deg(u_2) = 3$ $\quad$ $\deg(v_2) = 3$
   $\deg(u_3) = 3$ $\quad$ $\deg(v_3) = 3$
   $\deg(u_4) = 3$ $\quad$ $\deg(v_4) = 3$
   $\deg(u_5) = 3$ $\quad$ $\deg(v_5) = 3$
   $\deg(u_6) = 3$ $\quad$ $\deg(v_6) = 3$
   $\deg(u_7) = 3$ $\quad$ $\deg(v_7) = 3$
   $\deg(u_8) = 3$ $\quad$ $\deg(v_8) = 3$
   $\deg(u_9) = 3$ $\quad$ $\deg(v_9) = 3$
   $\deg(u_{10}) = \frac{3}{30}$ $\quad$ $\deg(v_{10}) = \frac{3}{30}$
4. The number of odd degrees in both graphs is same.
5. $f(u_1) = v_1$ $\quad$ $f(u_6) = v_8$
   $f(u_2) = v_9$ $\quad$ $f(u_7) = v_7$
   $f(u_3) = v_4$ $\quad$ $f(u_8) = v_5$
   $f(u_4) = v_3$ $\quad$ $f(u_9) = v_{10}$
   $f(u_5) = v_2$ $\quad$ $f(u_{10}) = v_6$

   So the above graphs are isomorphic.

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44.



1. No of vertices is same
2. No of edges also same.
3. $\deg(u_1) = 5$ $\deg(v_1) = 5$
   $\deg(u_2) = 5$ $\deg(v_2) = 5$
   $\deg(u_3) = 5$ $\deg(v_3) = 5$
   $\deg(u_4) = 5$ $\deg(v_4) = 5$
   $\deg(u_5) = 5$ $\deg(v_5) = 5$
   $\deg(u_6) = 5$ $\deg(v_6) = 5$
   $\deg(u_7) = 5$ $\deg(v_7) = 5$
   $\deg(u_8) = 5$ $\deg(v_8) = 5$

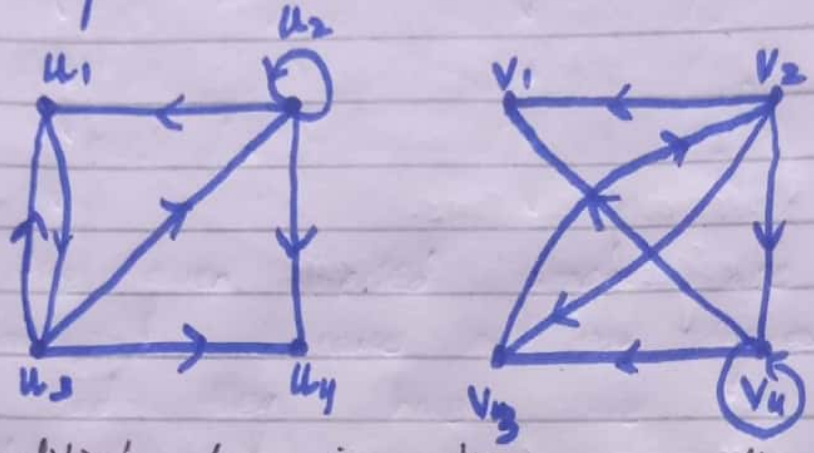
   40 40
4. No of odd degrees is same and there is no even degree.
5. $f(u_1) = v_1$ $f(u_6) = v_6$
   $f(u_2) = v_2$ $f(u_7) = v_7$
   $f(u_3) = v_3$ $f(u_8) = v_8$
   $f(u_4) = v_4$
   $f(u_5) = v_5$

So, the given graphs are isomorphic.

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In Exercise 61-64 determine whether the given directed graphs are isomorphic.

61.



Condition's for isomorphism in directed graphs:-

1. No of edges must be equal in both graphs.
2. No of vertices also must be equal.
3. In, out degree of all the vertices of G_1 must be present in In, out degree of G_2 (vertice).
4. But if 2 vertices of 0 In or out degree in both graphs they are not isomorphic.

Solution:-

1. No of edges in $G_1 = 7 =$ No of edges in G_2.
2. No of vertices in $G_1 = 4 =$ No of vertices in G_2.
3. formula:-

$$G_1(In, out) = G_2(In, out)$$

(one of In, out of G_1 has one to one function with G_2.)

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G_1 (In, out)

$\deg(u_1) = (2,1)$

$\deg(u_2) = (2,3)$

$\deg(u_3) = (1,3)$

$\deg(u_4) = (2,0)$

G_2 (In, out)

$\deg(v_1) = (2,0)$

$\deg(v_2) = (1,3)$

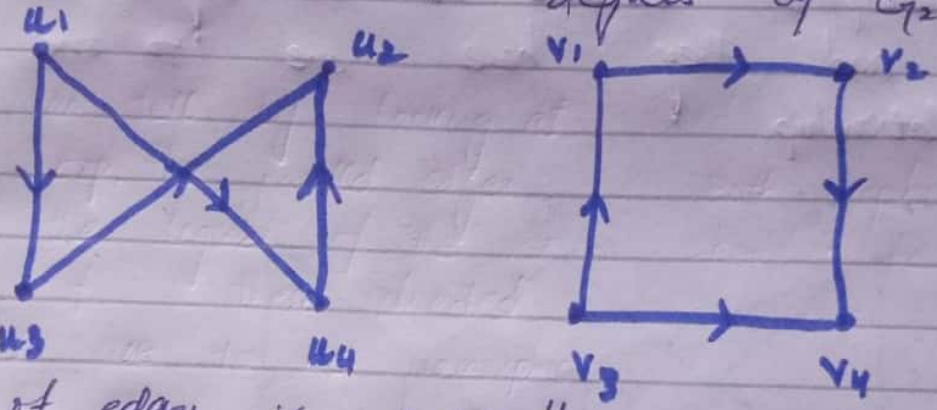
$\deg(v_3) = (2,1)$

$\deg(v_4) = (2,3)$

In this directed only out degree is 0.

So, the given graphs are isomorphic bcs all the degrees of G_1 have one to one relation with degrees of G_2.

62.



1. No of edges is same = 4
2. No of vertices is same = 4
3. $\deg(u_1) = (0,2)$ No bcs two of it's degrees are zero $\deg(v_1) = (1,1)$

   $\deg(u_2) = (2,0)$ $\deg(v_2) = (1,1)$

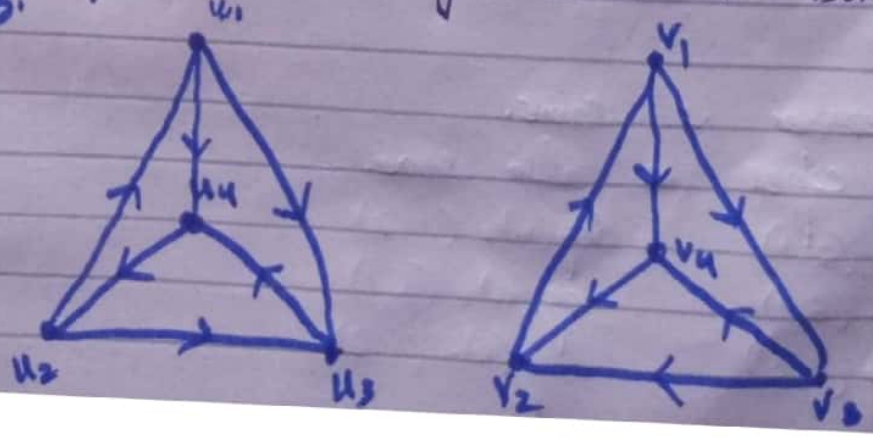
   $\deg(u_3) = (1,1)$ $\deg(v_3) = (0,2)$

   $\deg(u_4) = (1,1)$ $\deg(v_4) = (2,0)$

   so no one to one function No isomorphic.

Since 2 vertices u_1 and u_2 and similarly v_3 and v_4 in G_2 have 0 in or out degree so they are not isomorphic.

63.

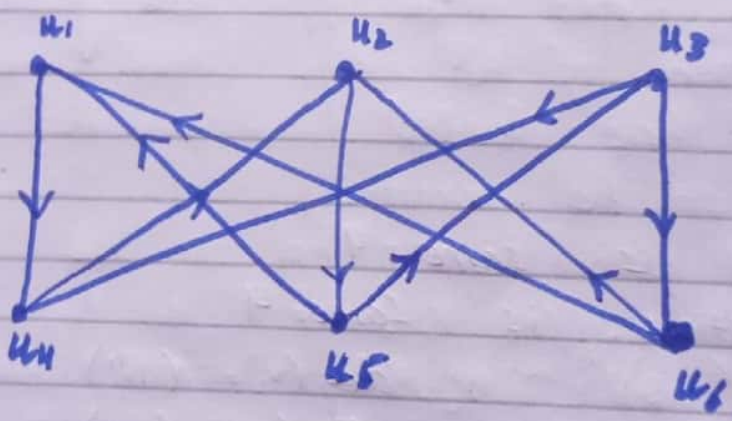


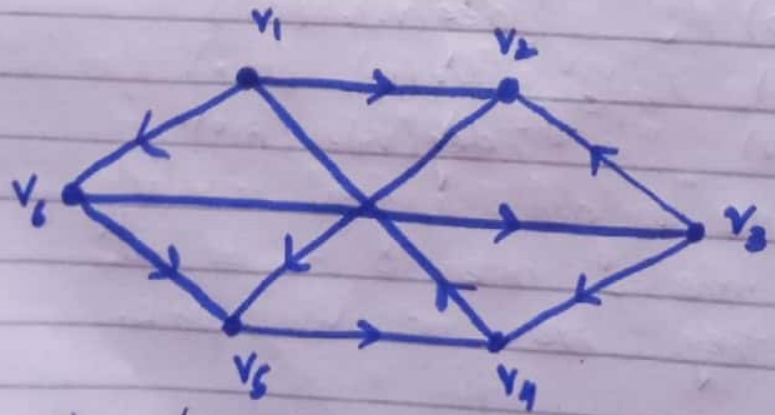
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1. No of vertices is same.
2. No of edges is same.
3. $\deg(u_1) = (1,2)$ — $\deg(v_1) = (1,2)$
   $\deg(u_2) = (1,2)$ — $\deg(v_2) = (2,1)$
   $\deg(u_3) = (2,1)$ — $\deg(v_3) = (1,2)$
   $\deg(u_4) = (2,1)$ — $\deg(v_4) = (2,1)$

So, the given directed graphs are isomorphic because no two degrees are same and they have one to one relationship.

64.





1. No of vertices is same.
2. No of edges is also same.
3. $\deg(u_1) = (2,1)$ — $\deg(v_1) = (1,2)$
   $\deg(u_2) = (2,1)$ — $\deg(v_2) = (2,1)$
   $\deg(u_3) = (1,2)$ — $\deg(v_3) = (1,2)$

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$\deg(u_4) = (2,1)$ $\quad\quad$ $\deg(v_4) = (2,1)$

$\deg(u_5) = (1,2)$ $\quad\quad$ $\deg(v_5) = (2,1)$

$\deg(u_6) = (1,2)$ $\quad\quad$ $\deg(v_6) = (1,2)$

So the given graphs are also isomorphic.



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Scanned with CamScanner

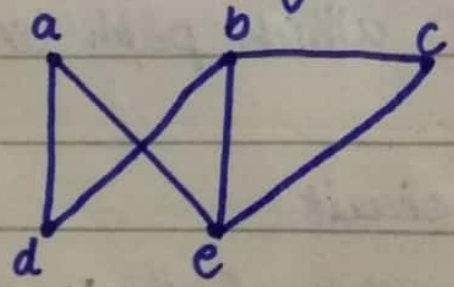
Graphs. 10.4

Does each of these lists of vertices form a path in the following graph?

(a) which paths are simple?

(b) which are circuit

(c) what are lengths of those that are paths?



(a) a, e, b, c, b

This is a path of length 4

But Not a simple path because the edge {b,c} is repeated

Not a circuit because it ends

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Scanned with CamScanner

with different vertex

(b) a, e, a, d, b, c, a

Not a path because there is no edge from c to a awording to given graph.

(c) e, b, a, d, b, e

This is not a path because there is no edge from b to a

(d) c, b, d, a, e, c

This a path of length 5

This is a simple path bcz no edge is repeated

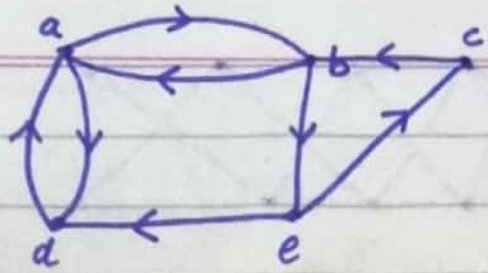
It is also a circuit because it is end with the same vertex.

Q#2: Does each of these list of vertices form a path in the following graph? Which path are simple

(b) Which are circuit

(c) What are lengths of those that are paths.

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(a) a, b, e, c, b

This is path of length of 4

Not a circuit

It is simple path bcz no edge is repeated

(b) a, d, a, d, a

This is path of length 4

which is a circuit

It is not a simple path. because edge is repeated more than once.

(c) a, d, b, e, a

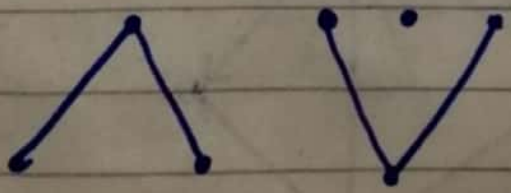
This is not a path bcz there is no edge from d to b.

(d) a, b, e, c, b, d, a

This is not a path because there is no edge from b to d.

3-5 Determine whether the given graph is connected.

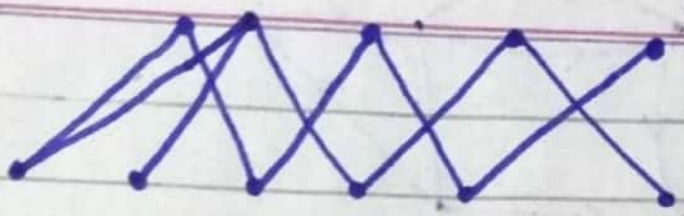
3



Not connected.

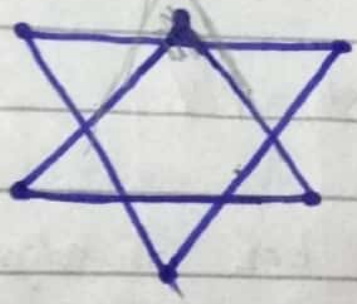
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4



connected.

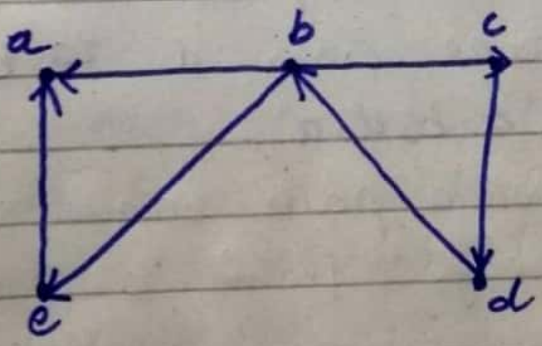
5



Not connected.

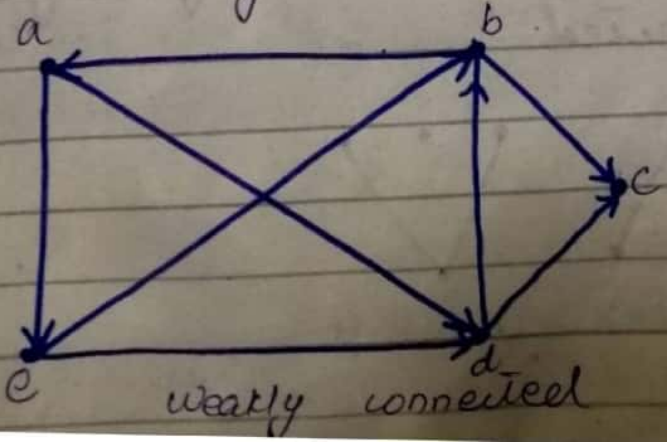
11// Determine whether each of these graphs is strongly connected and if not, whether it is weakly connected

(a)



weakly connected

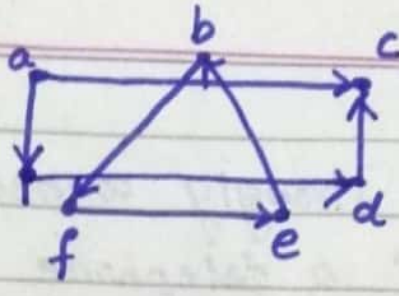
(b)



weakly connected

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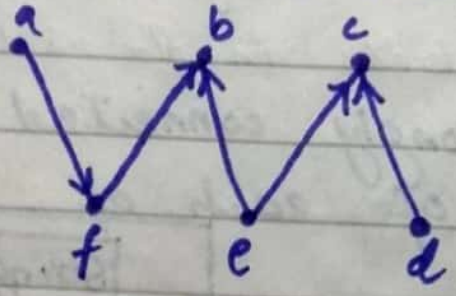
(c)



neither strongly or weakly connected

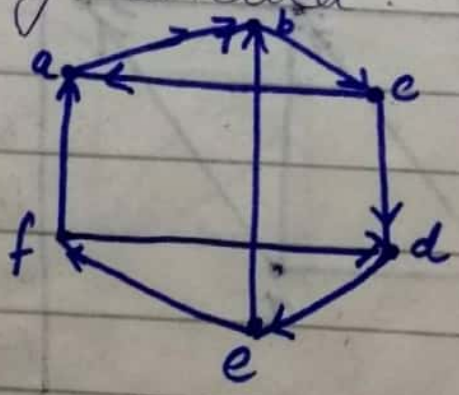
12 Determine whether each of these graphs is strongly connected and if not, whether it is weakly connected

(a)



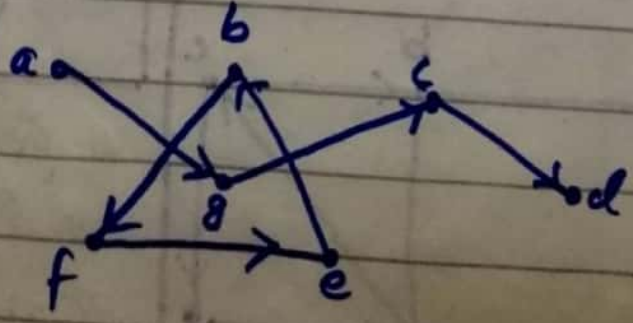
weakly connected

(b)



Strongly connected

(c)



Neither weakly or strongly connected

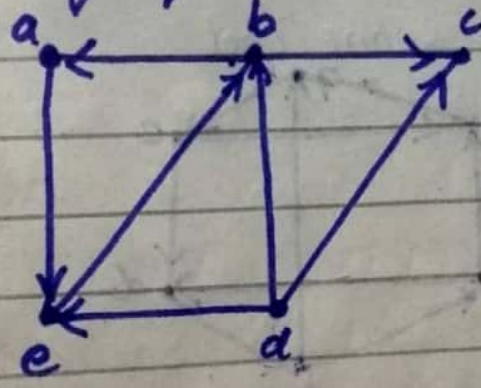
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Hint

3 What do the strongly connected components of a telephone cell graph represent

The maximum sets of phone numbers for which it is possible to find directed path between every two different numbers in the set.

4 Find the strongly connected components of each of these graphs

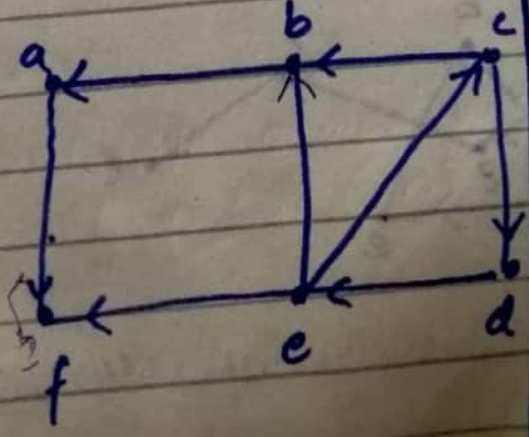


Strongly connected graph

A graph is said to be strongle connected graph if every pair of vertice in the graph contain a path between each other.

{ a, e, b }, { c }, { d }

(b)



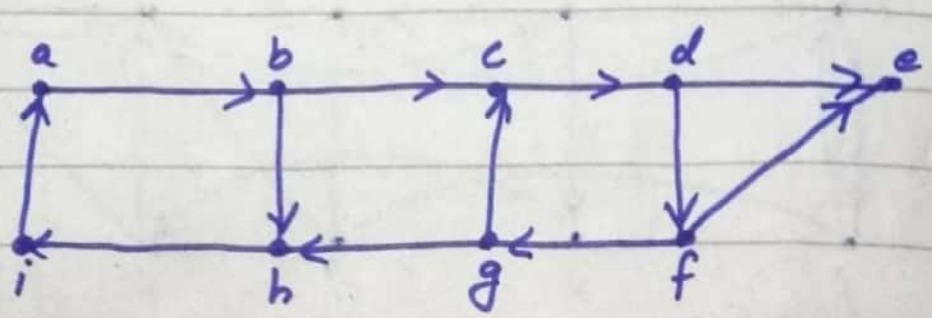
Weakly connected graph

A graph is said to be weakly connected graph if there doesn't exist any path b/w two pair of vertice

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$\{a\}, \{b\}, \{f\}, \{c, d, e\}$

(c)

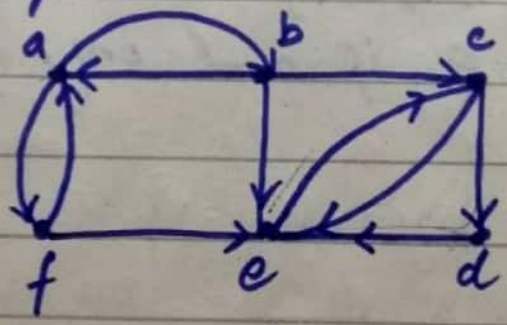


~~$\{a, b, h, i\}, \{c, d, g, f\}, \{e\}$~~

$\{a, b, c, d, f, g, h, i\}, \{e\}$

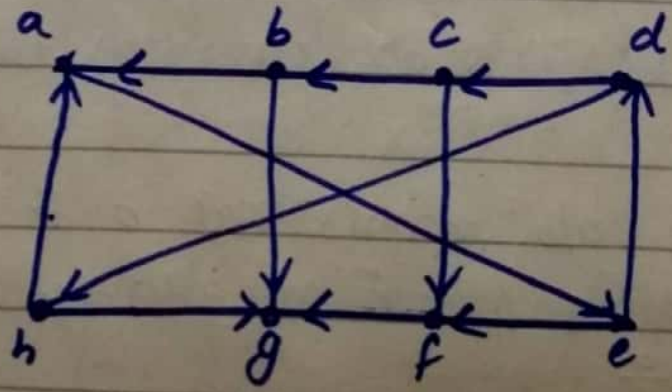
15 Find the strongly connected components of each of these graphs

(a)



$\{a, b, f\}, \{c, d, e\}$

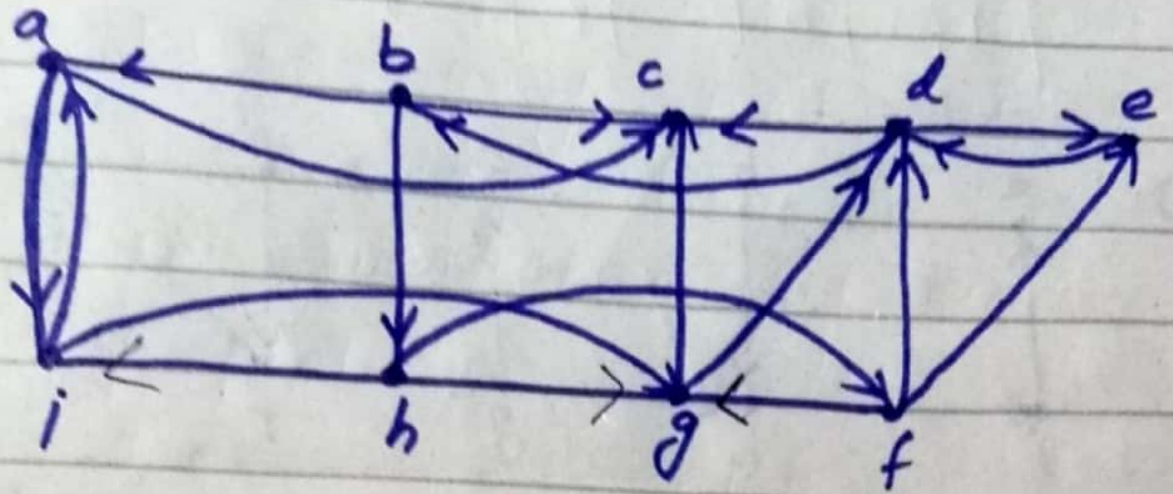
(b)



$\{a, e, d, c, b, h\}, \{g\}, \{f\}$

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c)



{a, b, c, d, e, f, g, h, i}, {c}

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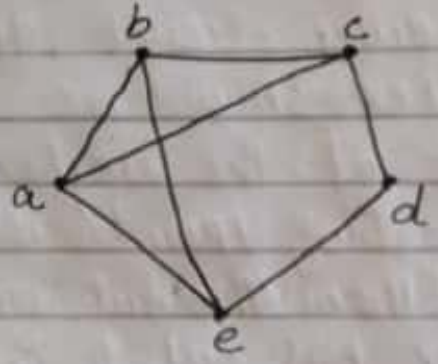
CHAPTER # 10:-

GRAPHS

EXERCISE 10.5

In Excercises 1-8, determine whether the given graph has an Euler circuit. Construct such a circuit when one exists. If no Euler circuit exists, determine whether the graph has an Euler path and construct such a path if one exists.

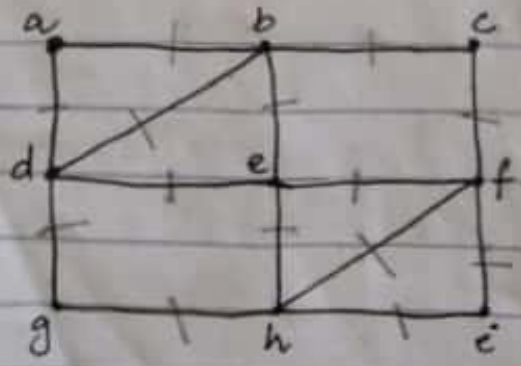
1.



deg(a) = 3
deg(b) = 3
deg(c) = 3
deg(d) = 4
deg(e) = 3

Niether

2.



deg(a) = 2
deg(b) = 4
deg(c) = 2
deg(d) = 4
deg(e) = 4
deg(f) = 4
deg(g) = 2
deg(h) = 4
deg(i) = 2

Euler circuit ; a, b, c, f, i, h, g, d, e, f, h, e, b, d, a

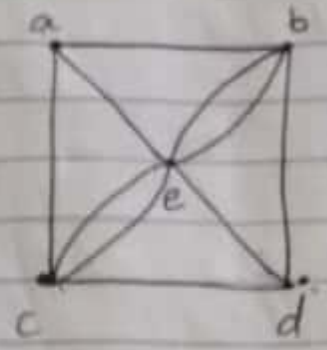
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UnDirected graph:-

* An Euler circuit is possible if every vertex has even degree.
* An Euler path is possible, but an Euler circuit is not possible, it exactly two vertices have odd degree.
* No Euler path is possible if more than two vertices have odd degree.

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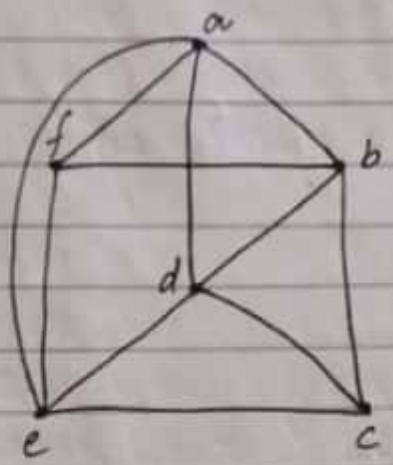
3.



deg(a) = 3
deg(b) = 4
deg(c) = 4
deg(d) = 3
deg(e) = 6

Euler path ; a, e, c, e, b, e, d, b, a, c, d

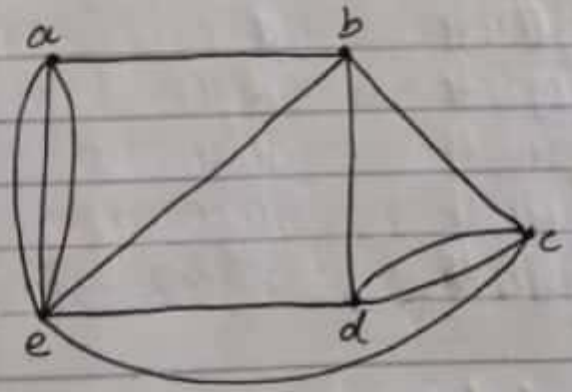
4.



deg(a) = 4
deg(b) = 4
deg(c) = 3
deg(d) = 4
deg(e) = 4
deg(f) = 3

Euler path ; f, a, b, c, d, e, f, b, d, a, e, c

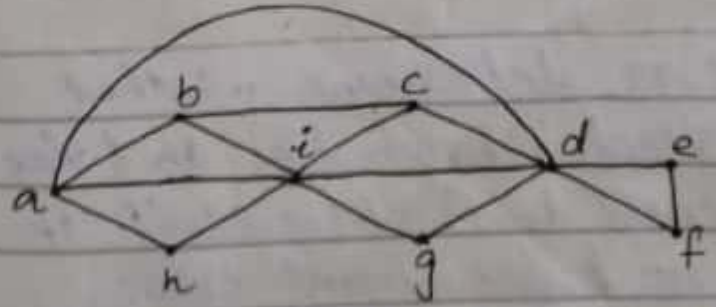
5.



deg(a) = 4
deg(b) = 4
deg(c) = 4
deg(d) = 4
deg(e) = 4

Euler circuit ; a, b, c, d, c, e, b, d, e, a, e, a

6.

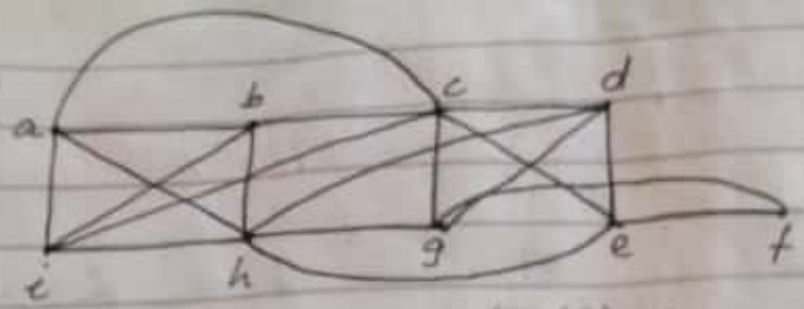


deg(a) = 4
deg(b) = 3
deg(c) = 3
deg(d) = 6
deg(e) = 2
deg(f) = 2
deg(g) = 2
deg(h) = 2
deg(i) = 6

Euler path ; b, c, d, e, f, d, g, i, d, a, h, i, a, b, i, c

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7.

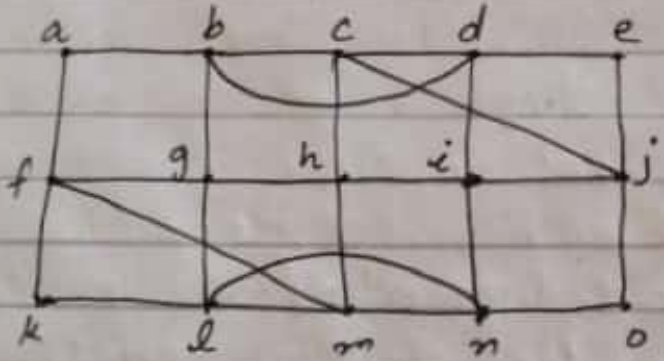


deg(a) = 4
deg(b) = 4
deg(c) = 6
deg(d) = 4
deg(e) = 4
deg(f) = 2
deg(g) = 4
deg(h) = 6
deg(i) = 4

Euler circuit

a, i, h, g, d, e, f, g, c, e, h, d, c, a, b, i, b, c, b, a

8.



deg(a) = 2
deg(b) = 4
deg(c) = 4
deg(d) = 4
deg(e) = 2
deg(f) = 4
deg(g) = 4
deg(h) = 4
deg(i) = 4
deg(j) = 4
deg(k) = 2
deg(l) = 4
deg(m) = 4
deg(n) = 4
deg(o) = 2

Euler circuit

a, b, c, d, e, j, c, h, i, d, b, g, h, m, n, o, j, i
n, l, m, f, g, l, k, f, a

In Exercises 18-23 determine whether the directed graph shown has an Euler circuit. Construct an Euler circuit if one exists. If no Euler circuit exist, determine whether the directed graph

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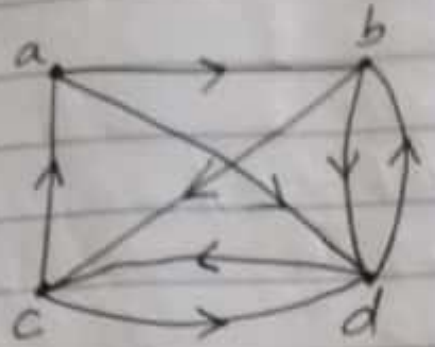
= 28

Directed Graph:-

* For every graph has an euler circuit, the d: in-degree of each vertex should be equal to the out-degree of the vertex.
* For Euler path, the starting point/vertex has in-degree smaller than the out-degree and ending point/vertex has out-degree smaller than the in-degree and the remaining vertices should must have the same degree.

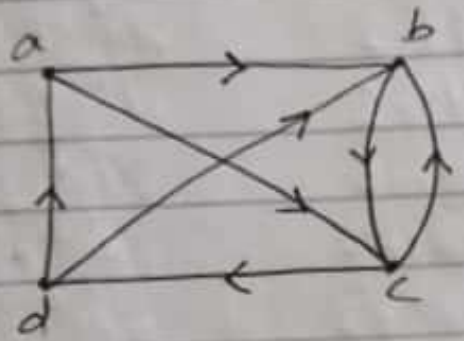
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has an Euler path. Construct an Euler path if one exists.



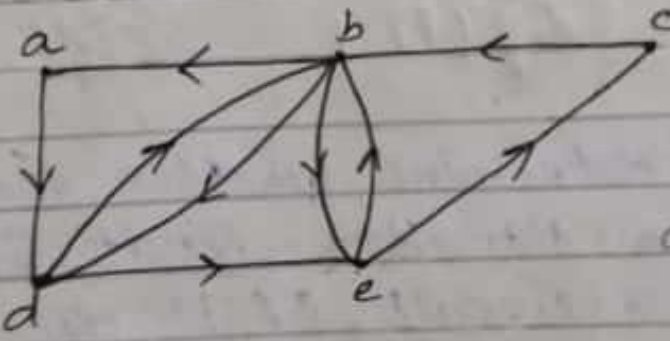
$\deg^-(a) = 1$ $\deg^+(a) = 2$
$\deg^-(b) = 2$ $\deg^+(b) = 2$
$\deg^-(c) = 2$ $\deg^+(c) = 2$
$\deg^-(d) = 3$ $\deg^+(d) = 2$

Euler path; a, b, d, b, c, d, c, a, d



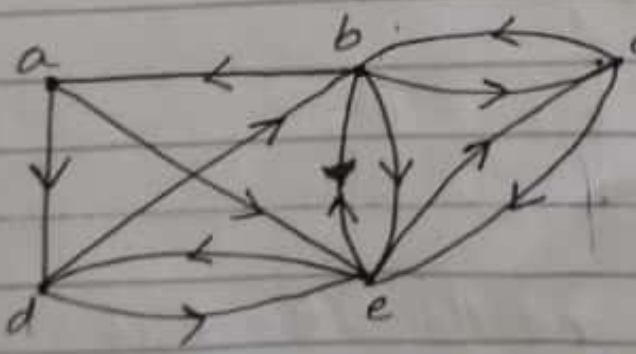
$\deg^-(a) = 1$ $\deg^+(a) = 2$
$\deg^-(b) = 3$ $\deg^+(b) = 1$
$\deg^-(c) = 2$ $\deg^+(c) = 2$
$\deg^-(d) = 1$ $\deg^+(d) = 2$

Niether a, b, c, b



$\deg^-(a) = 1$ $\deg^+(a) = 1$
$\deg^-(b) = 3$ $\deg^+(b) = 3$
$\deg^-(c) = 1$ $\deg^+(c) = 1$
$\deg^-(d) = 2$ $\deg^+(d) = 2$
$\deg^-(e) = 2$ $\deg^+(e) = 2$

Euler circuit; a, d, b, e, b, d, e, c, b, a

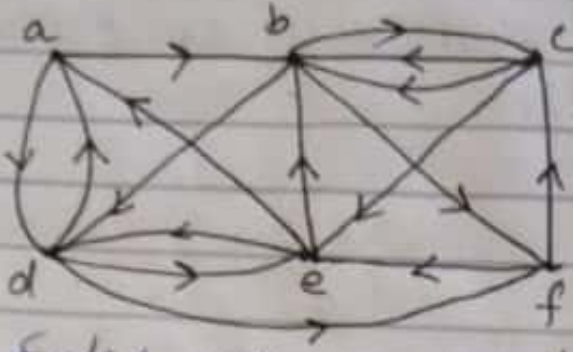


$\deg^-(a) = 1$ $\deg^+(a) = 2$
$\deg^-(b) = 3$ $\deg^+(b) = 3$
$\deg^-(c) = 2$ $\deg^+(c) = 2$
$\deg^-(d) = 2$ $\deg^+(d) = 2$
$\deg^-(e) = 4$ $\deg^+(e) = 3$

Euler path; a, d, e, d, b, a, e, c, e, b, c, b, e

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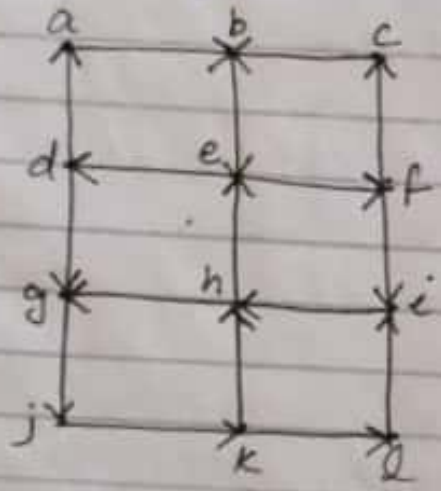
22.



$\deg^-(a) = 2$ $\deg^+(a) = 2$
$\deg(b) = 4$ $\deg^+(b) = 3$
$\deg(c) = 2$ $\deg^+(c) = 3$
$\deg(d) = 3$ $\deg^+(d) = 3$
$\deg(e) = 3$ $\deg^+(e) = 3$
$\deg(f) = 2$ $\deg^+(f) = 2$

Euler path; c, e, b, d, c, b, f, d, e, f, e, a, f, a, b, c, b

23.

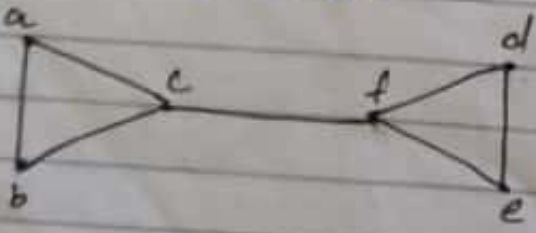


$\deg^-(a) = 1$ $\deg^+(a) = 1$
$\deg^-(b) = 2$ $\deg^+(b) = 1$
$\deg^-(c) = 1$ $\deg^+(c) = 1$
$\deg^-(d) = 1$ $\deg^+(d) = 1$
$\deg^-(e) = 2$ $\deg^+(e) = 2$
$\deg^-(f) = 1$ $\deg^+(f) = 2$
$\deg^-(g) = 2$ $\deg^+(g) = 1$
$\deg^-(h) = 2$ $\deg^+(h) = 2$
$\deg^-(i) = 2$ $\deg^+(i) = 1$
$\deg^-(j) = 1$ $\deg^+(j) = 1$
$\deg^-(k) = 1$ $\deg^+(k) = 2$
$\deg^-(l) = 1$ $\deg^+(l) = 1$

Niether

In Exercises 30-36 determine whether the given graph has a Hamilton circuit. If it does, find such a circuit, If it does not give an argument to show why no such circuit exists.

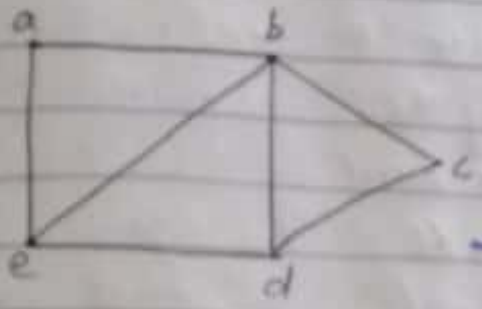
30.



This graph can have no Hamilton circuit b/c of the cut edge $\{c, f\}$. Every simple circuit must be confined to one of the two obtained by deleting this edge.

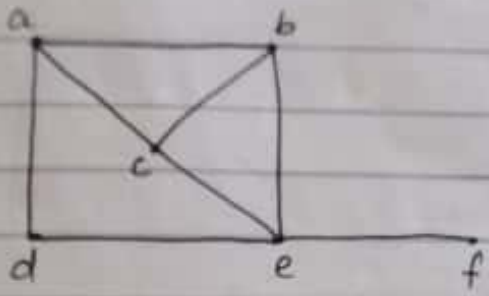
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31.



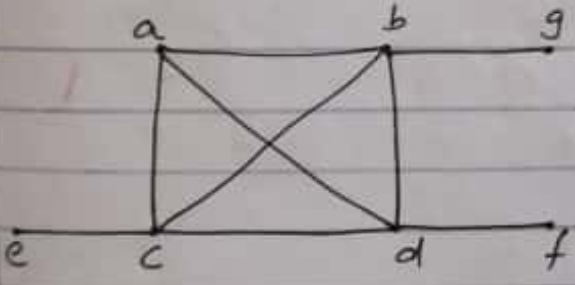
Hamilton circuit ; a, b, c, d, e, a

32.



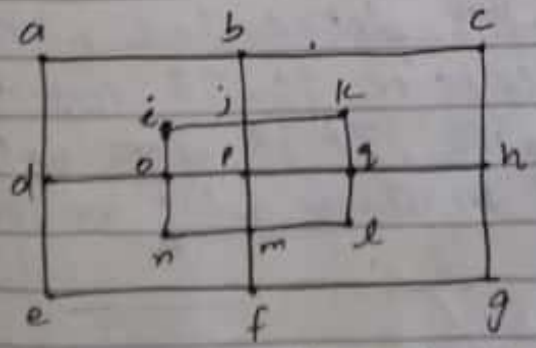
This graph can have no Hamilton circuit because of cut edge {e, f}.

33.



This graph has no Hamilton circuit because once a purposed circuit has reached e it would have nowhere to go.

34.

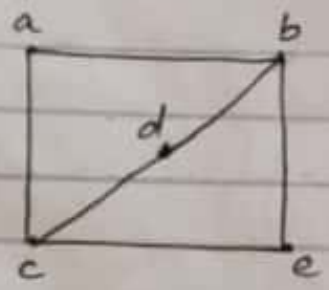


No Hamilton circuit. If it did, then certainly the circuit would have to contain edges {d, a}

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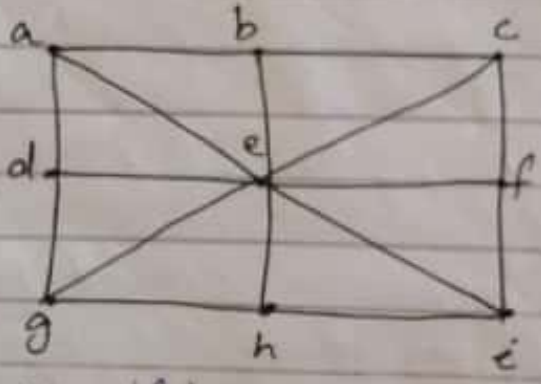
and $\{a,b\}$, since these are the only edges incident to vertex a. By the same reasoning, the circuit would have to contain the other six edges around the outside of the figure. These eight edges already complete a circuit, and this circuit omits the 9 vertices on the inside

35.



No Hamilton circuit, because every edge in the graph is incident to a vertex of degree 2 and therefore must be in a vertex.

36.



Hamilton circuit ; a, d, g, h, i, f, c, e, b, a

In Excercises 37-43 determine whether the graphs in Excercise 30-36 respectively have a Hamilton path. If it does not, give an argument to show why no such path exist.

37. a, b, c, f, d, e is a Hamilton path.

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38. a, b, c, d, e is Hamilton path.

39. f, e, d, a, b, c is Hamilton path.

40. This graph has no Hamilton path. There are three verbices of degree 1; each of them would have to be an end verbex of every Hamilton path. Since a path has only 2 ends, this is impossible.

41. No Hamilton path exists. There are 8 vertices of degree 2, and only two of them can be end vertices of path. For each of other six, their two incident edges must be in the path.

42. d, c, a, b, e is Hamilton path.

43. a, b, c, f, i, h, g, d, e is a Hamilton path.

Method 1:-

To check the if given graph is Hamilton Circuit or not Conditions are-

1. Starting and ending point is same
2. every vester is visited exactly once.

Method 2:-

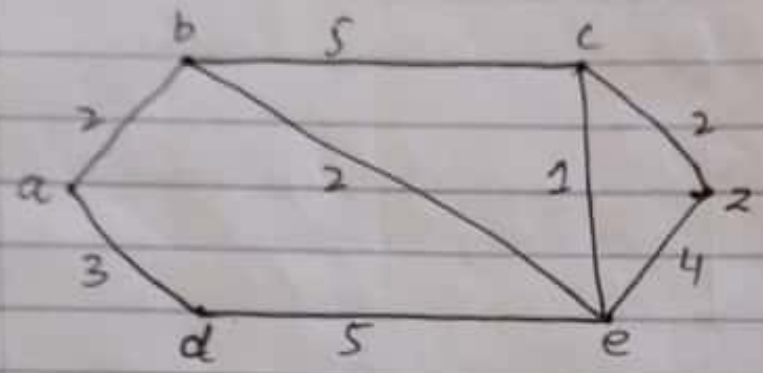
1. Count no of vertices. denoted by n.
2. Find n/2.
3. Then number degrees of each vertices must be greater than or equal to answer of n/2. If less It is not Hamilton circuit.

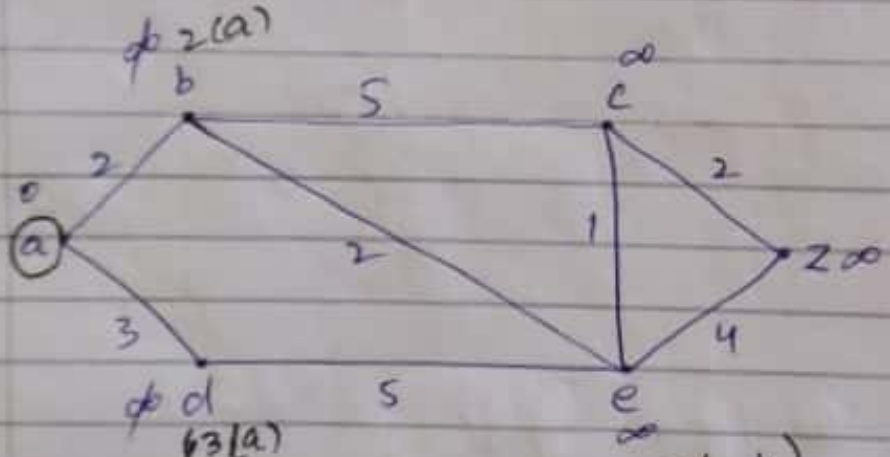
Scanned with CamScanner

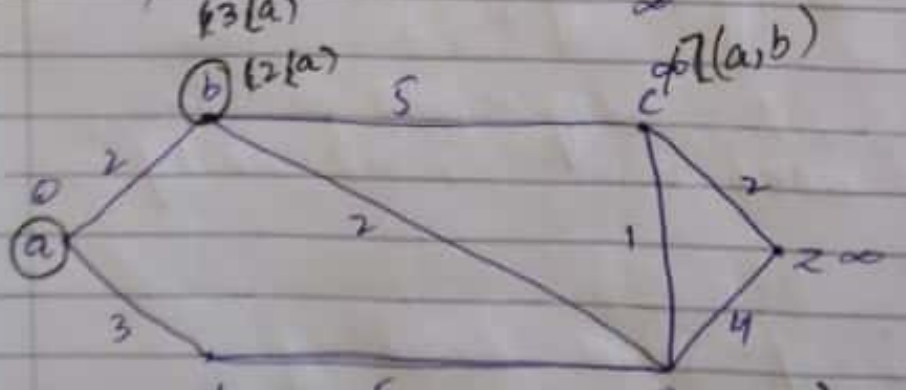
Exercise 10.6

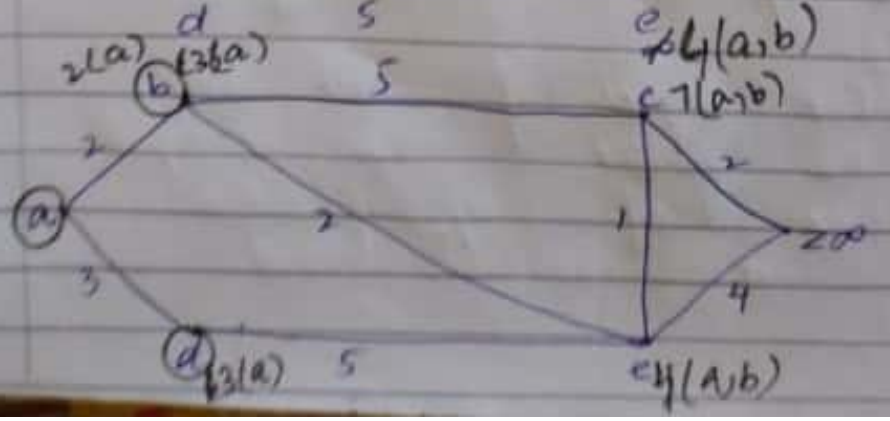
In Excercise 2-4, find the length of a shortest path between a and z in the given weighted graph.

2.

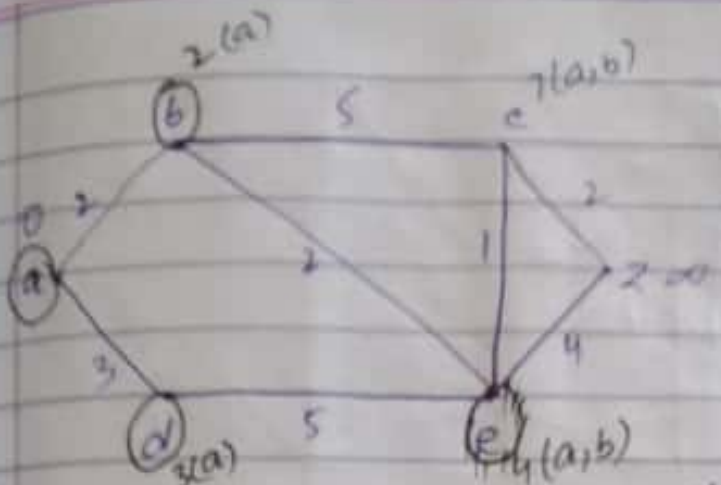


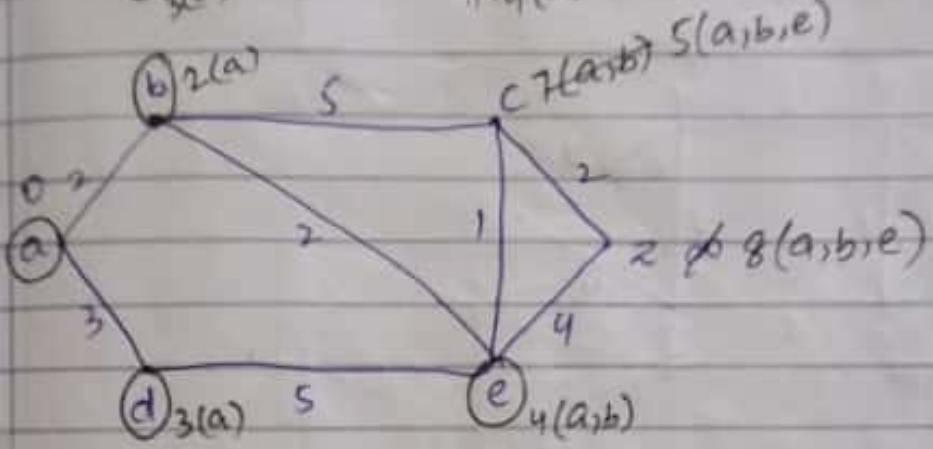


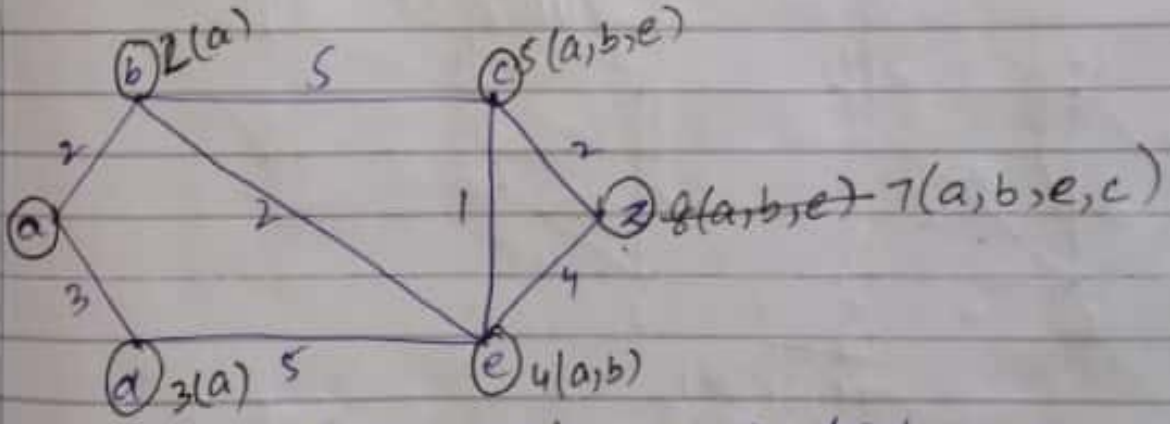




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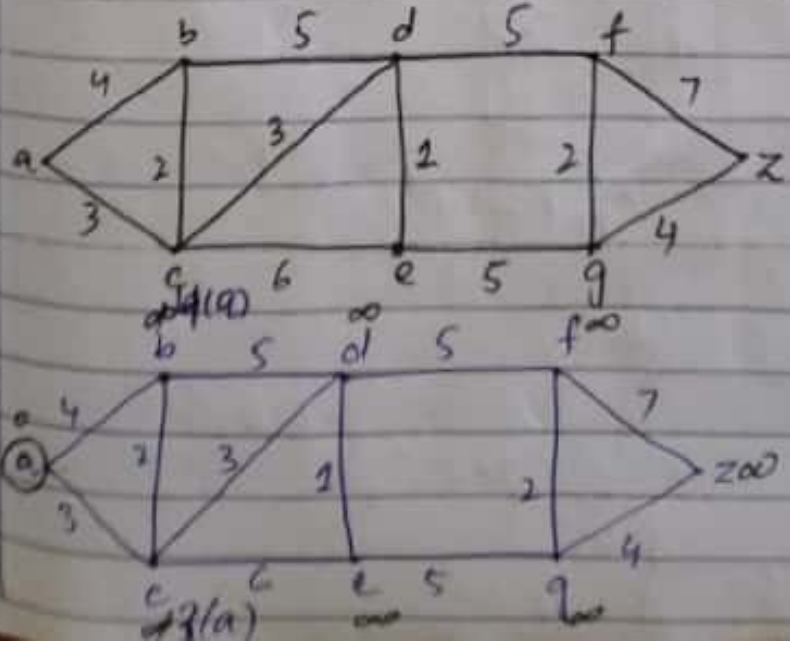


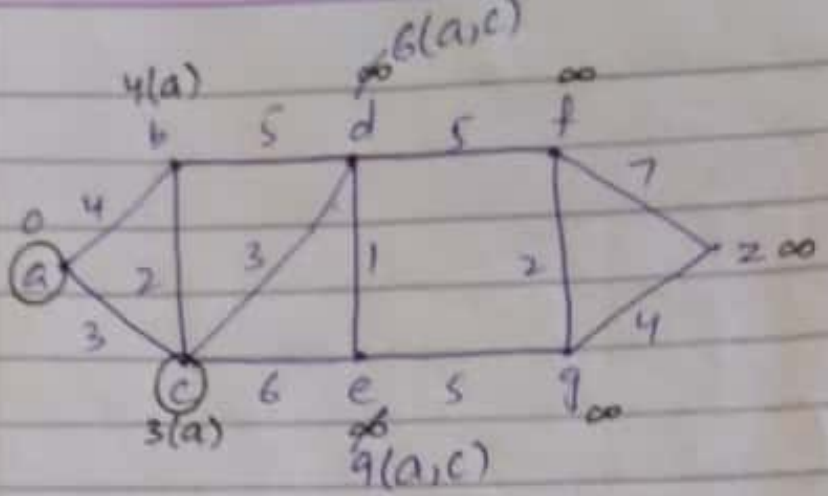


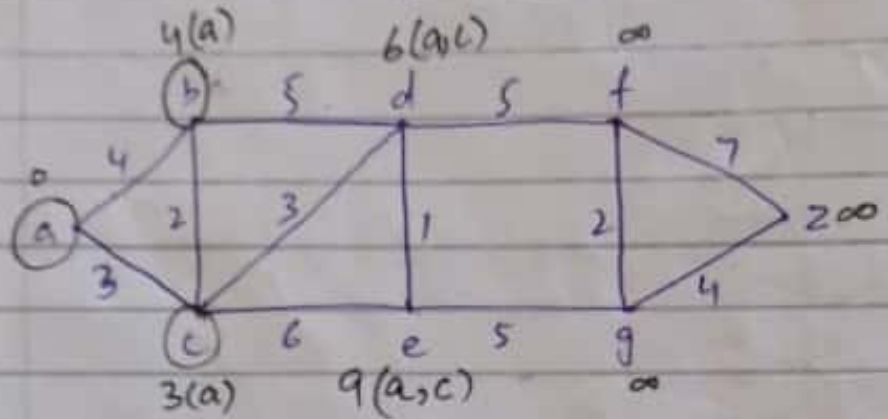


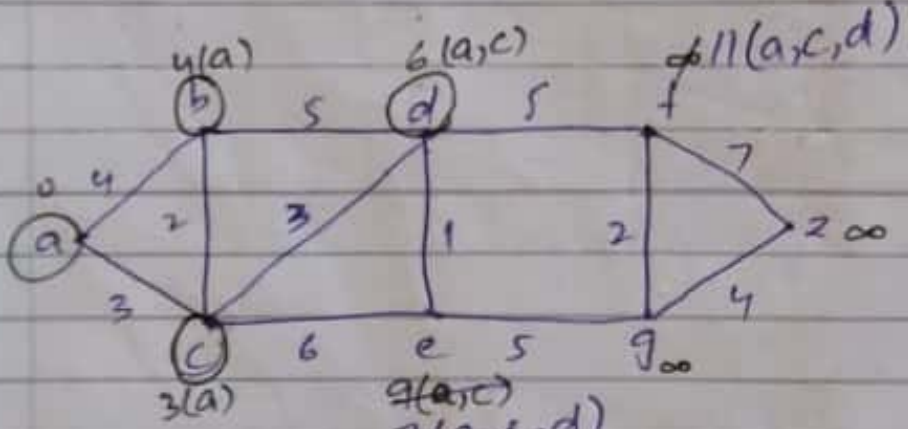
a, b, e, c, z is a shortest path whose length is 7.

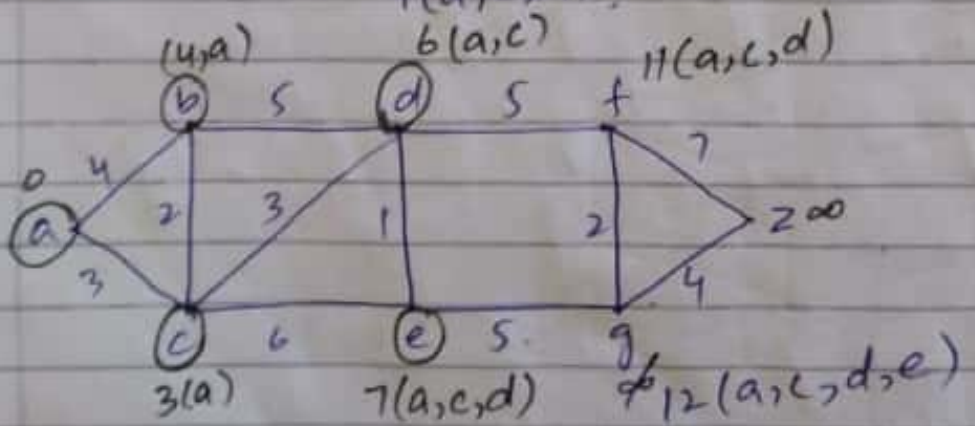
3.

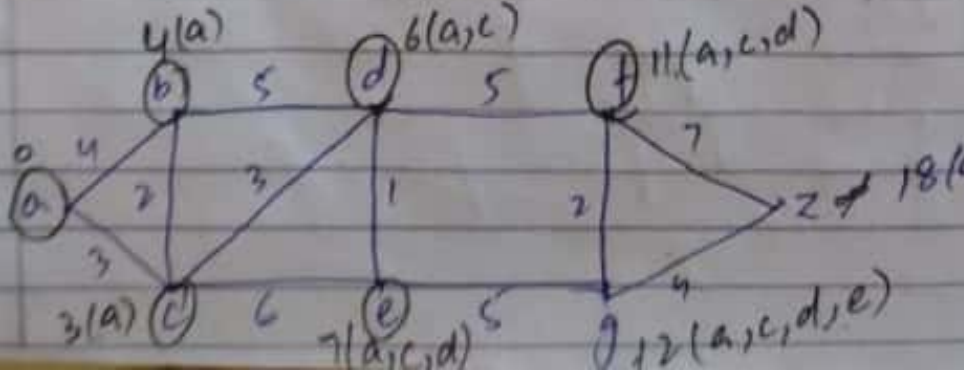




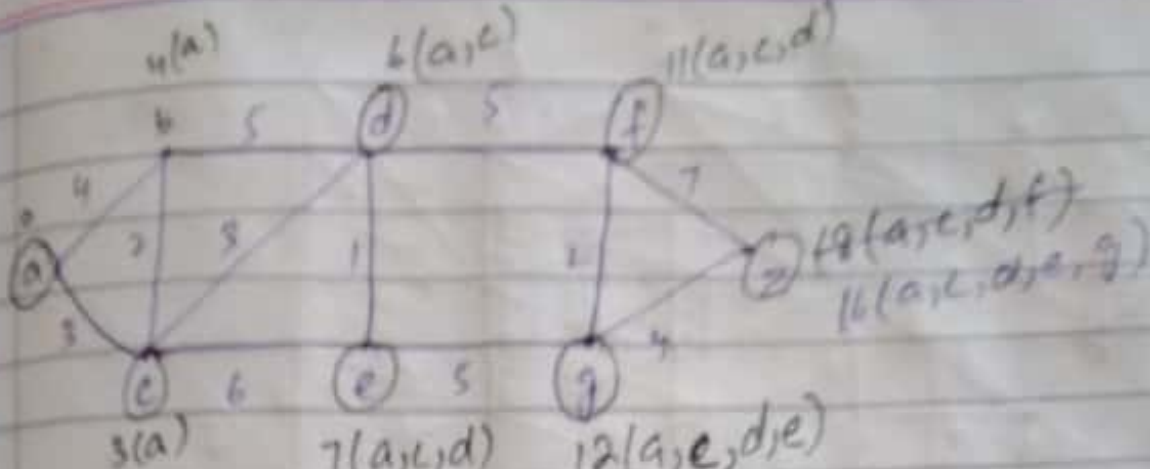






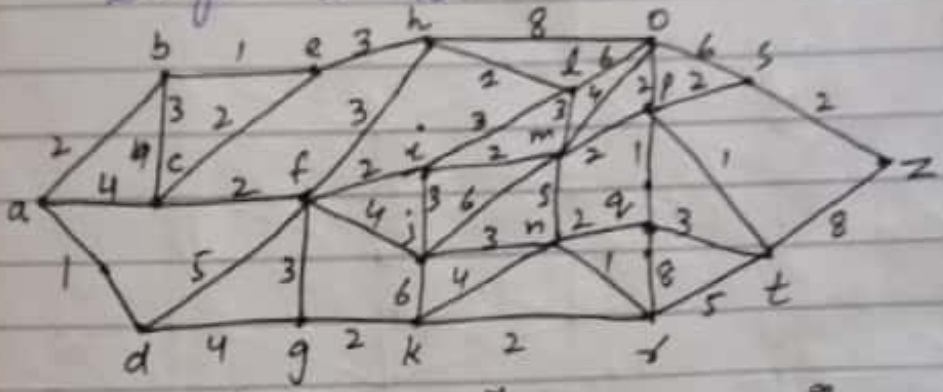


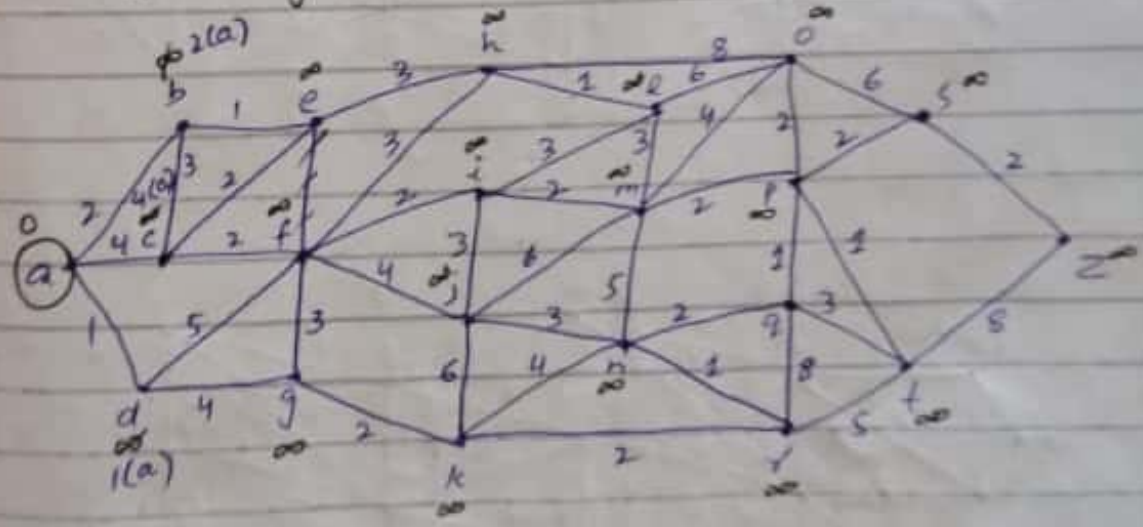
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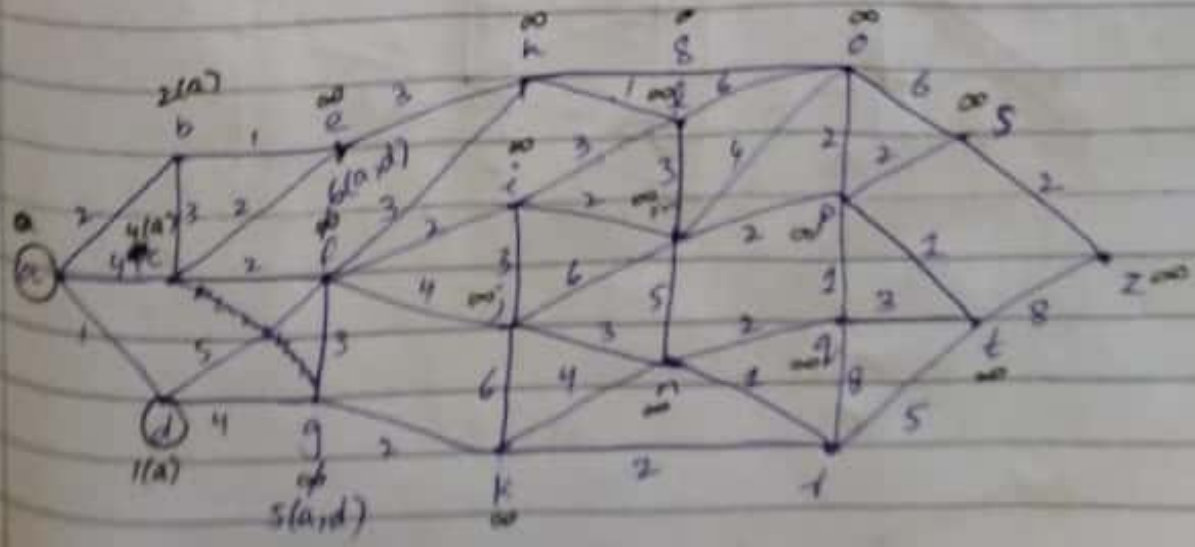


a, c, d, e, g, z is a shortest path whose length is 16.

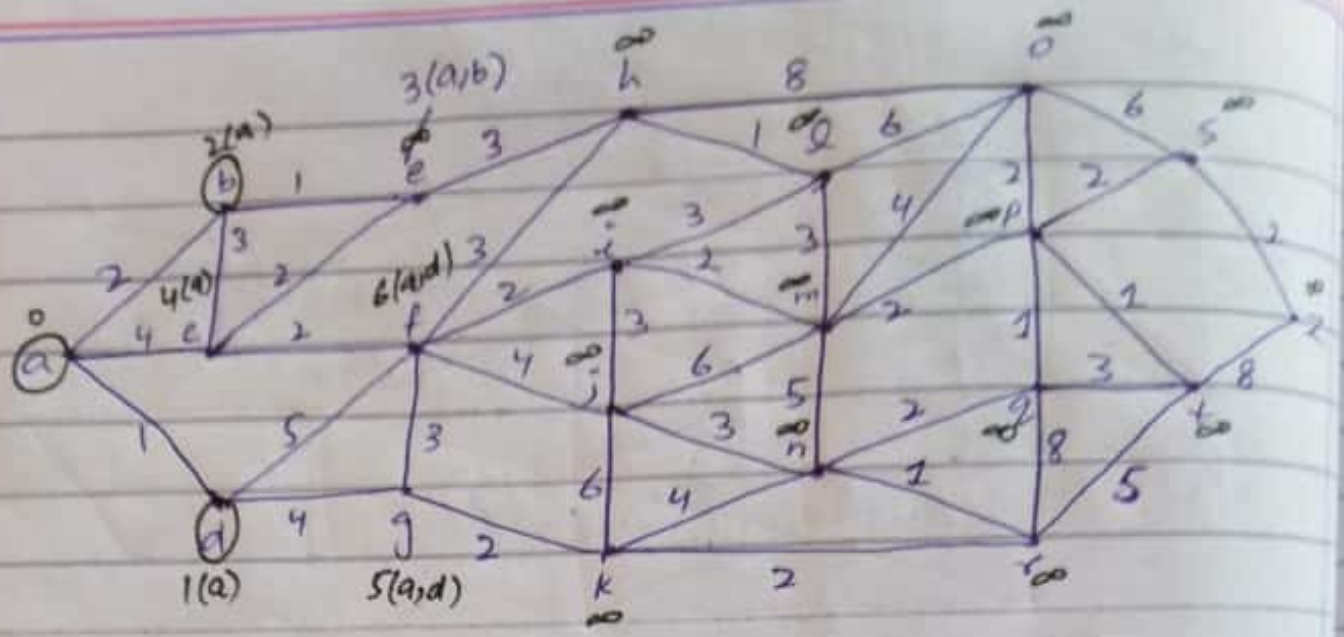
4.

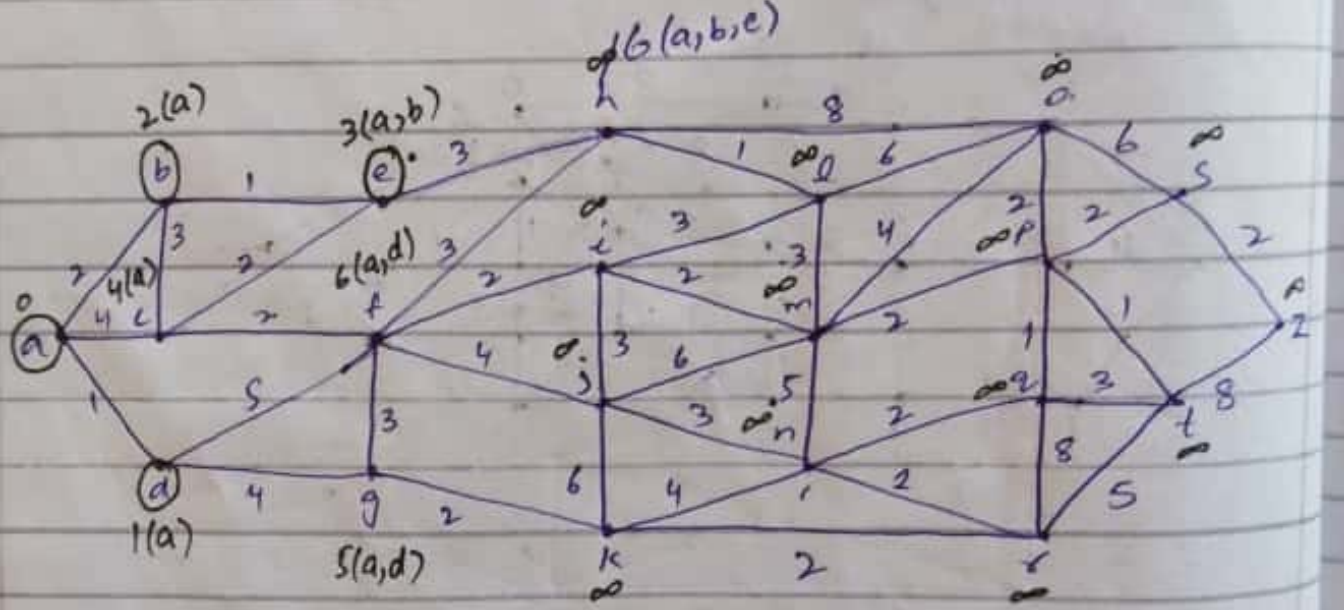


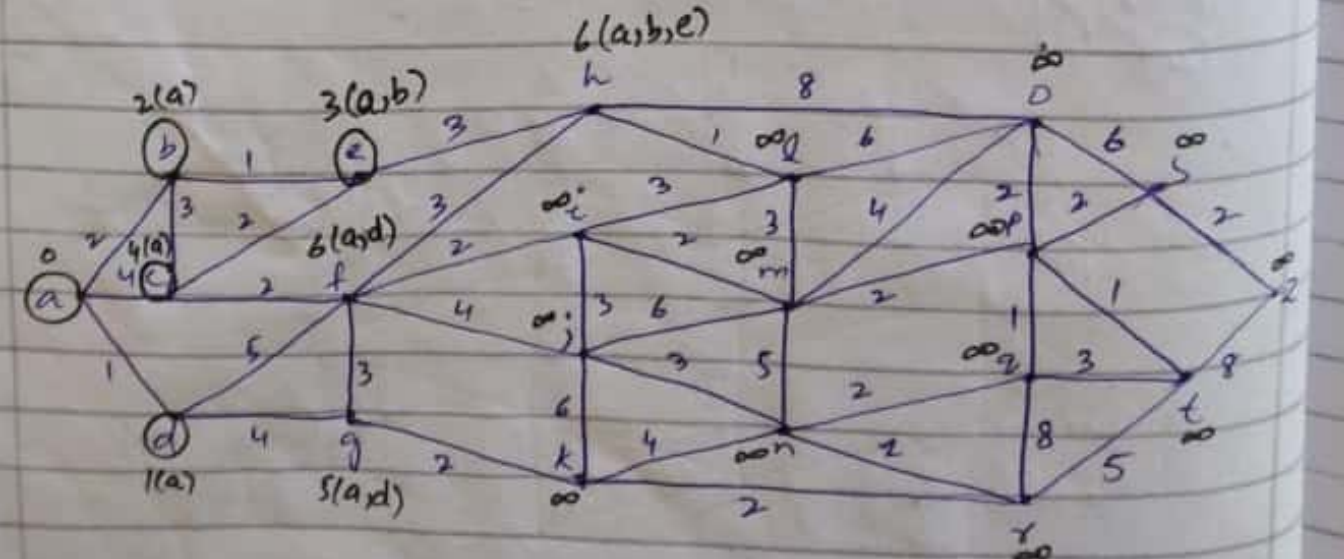




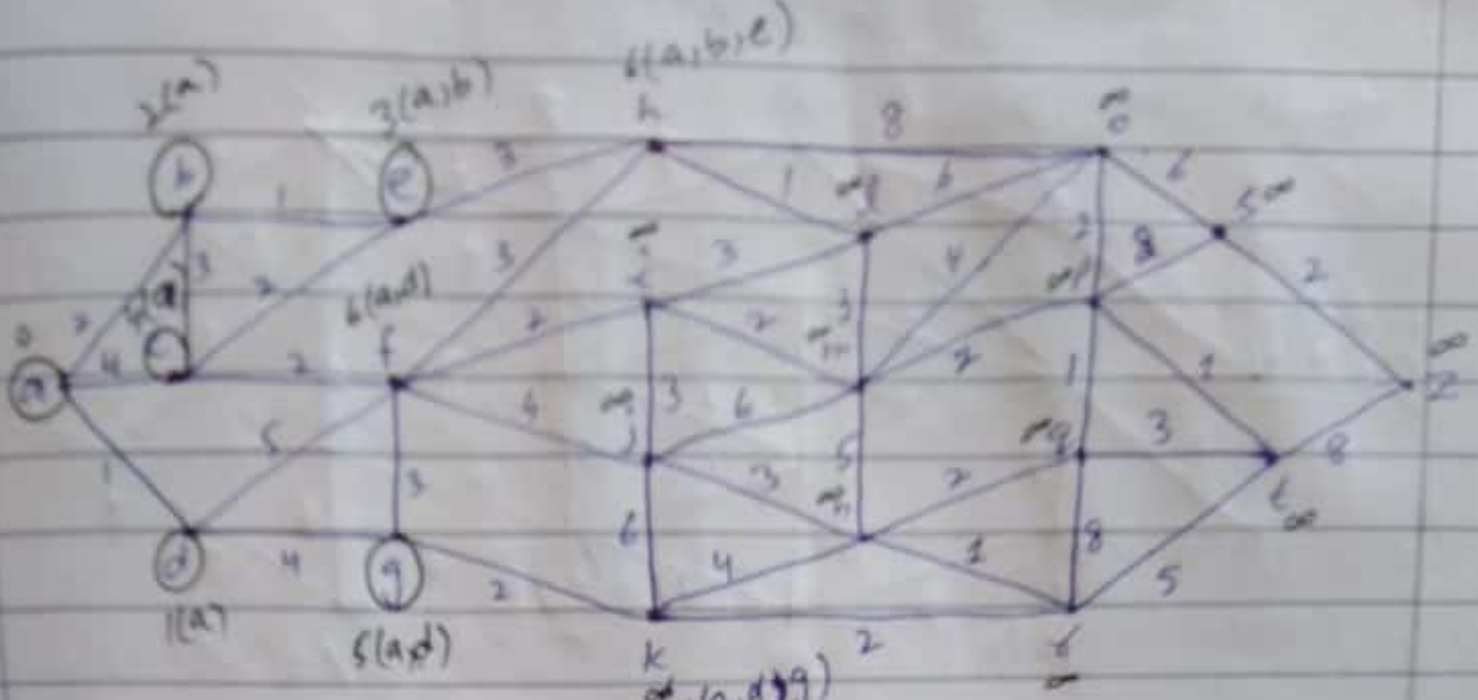
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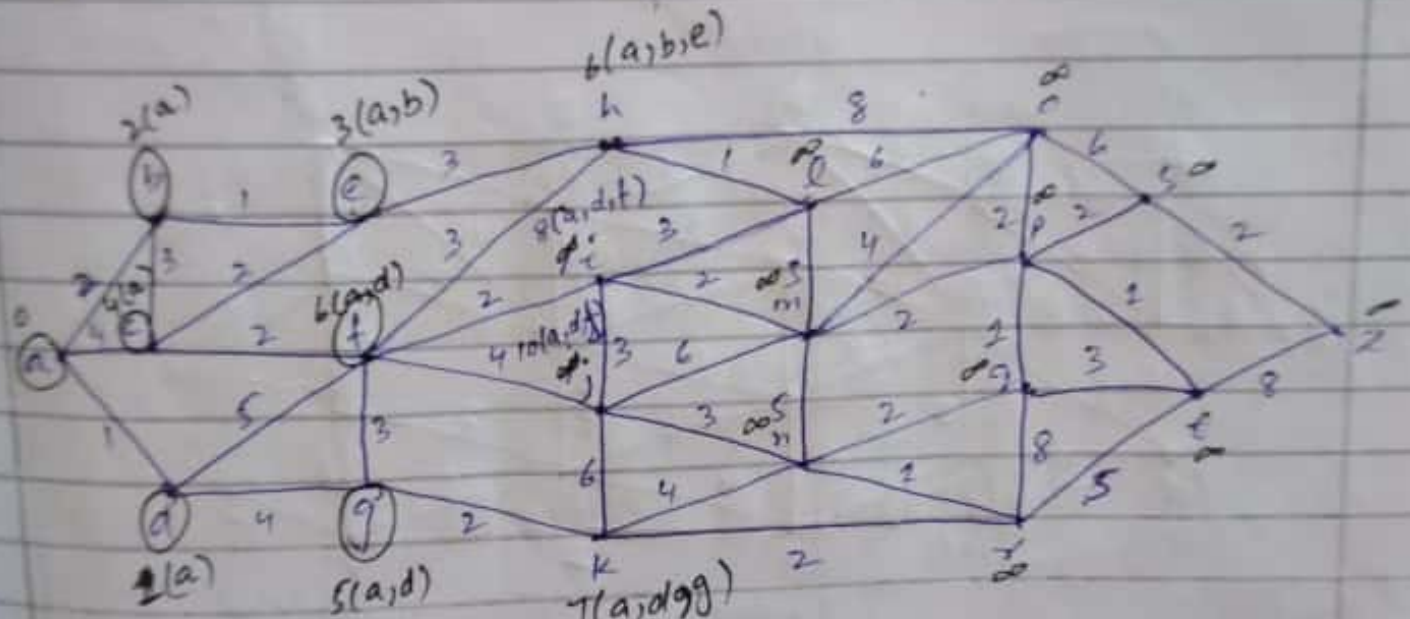


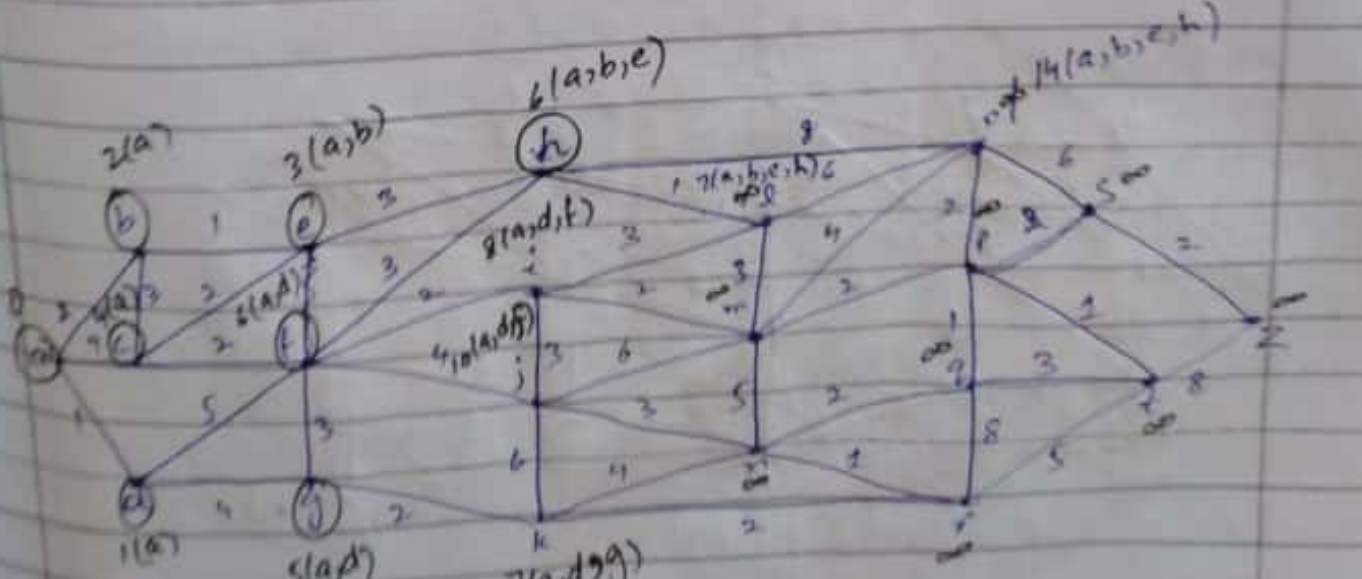




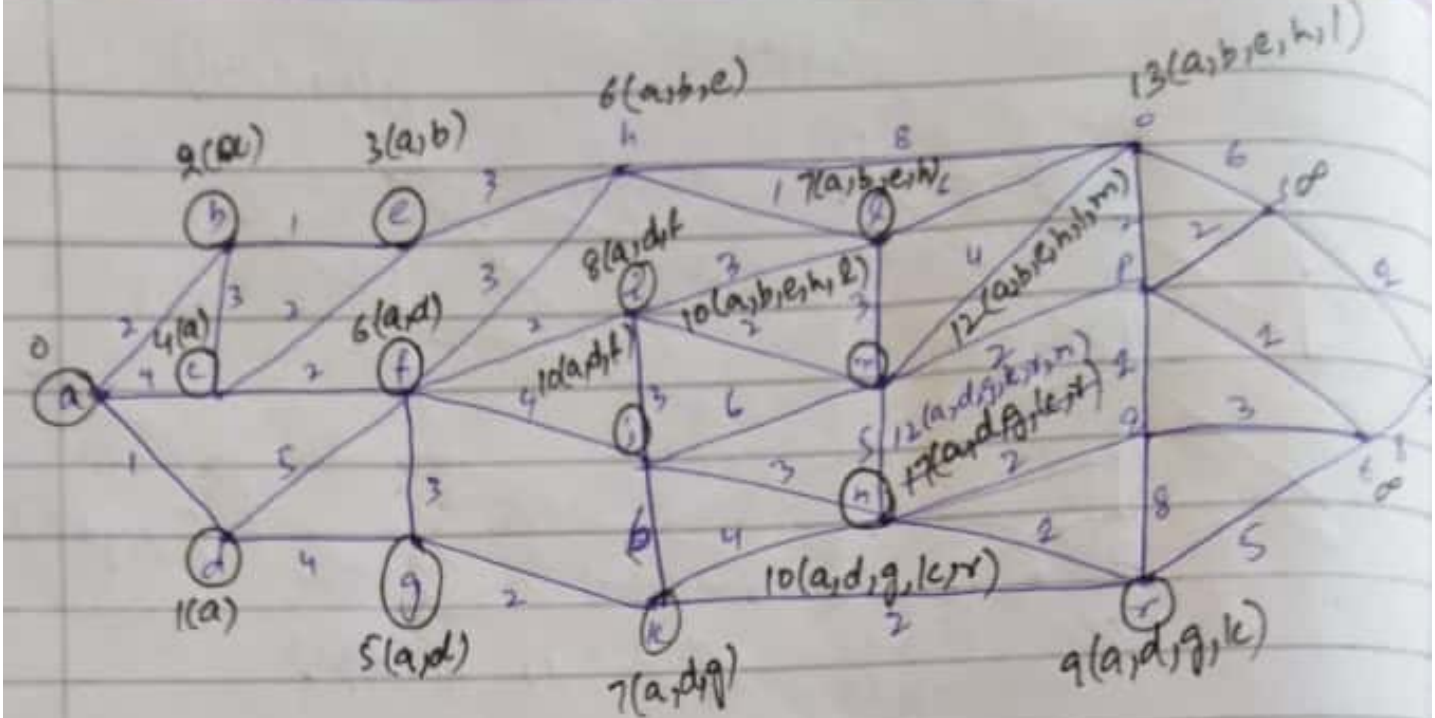
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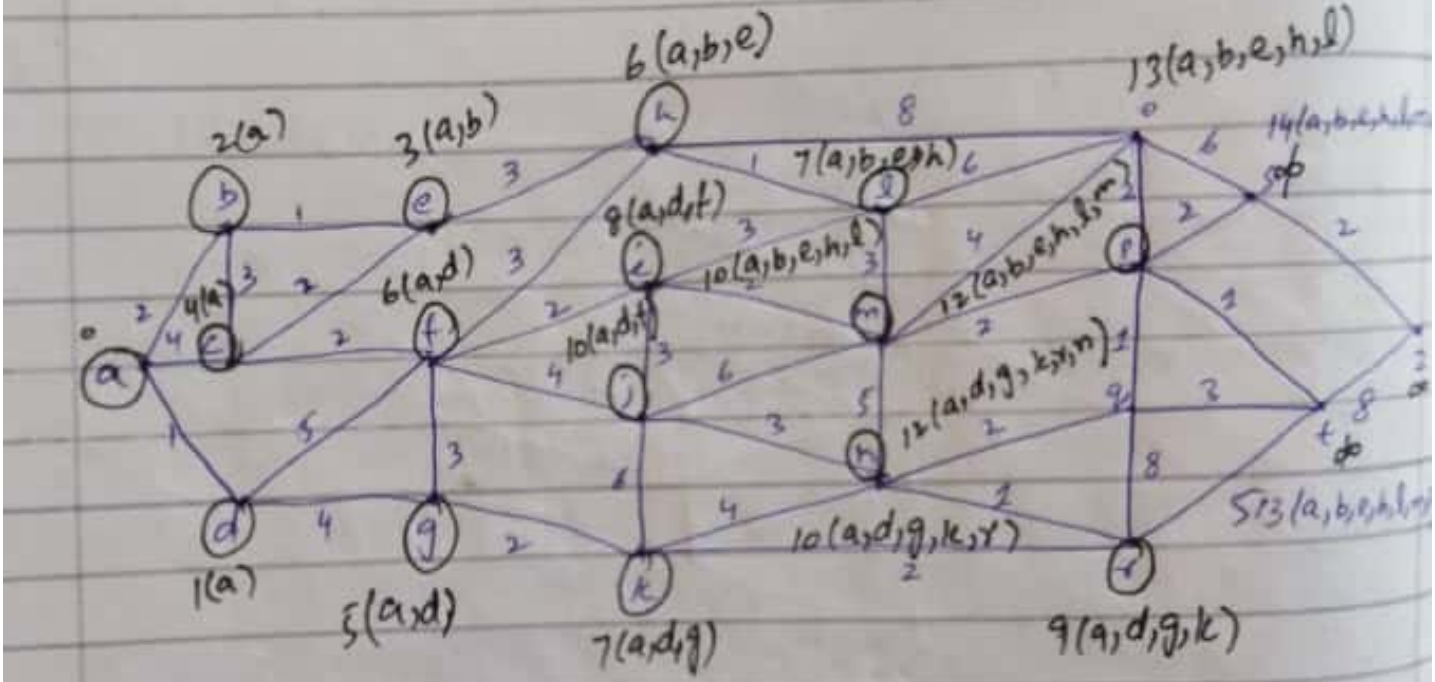


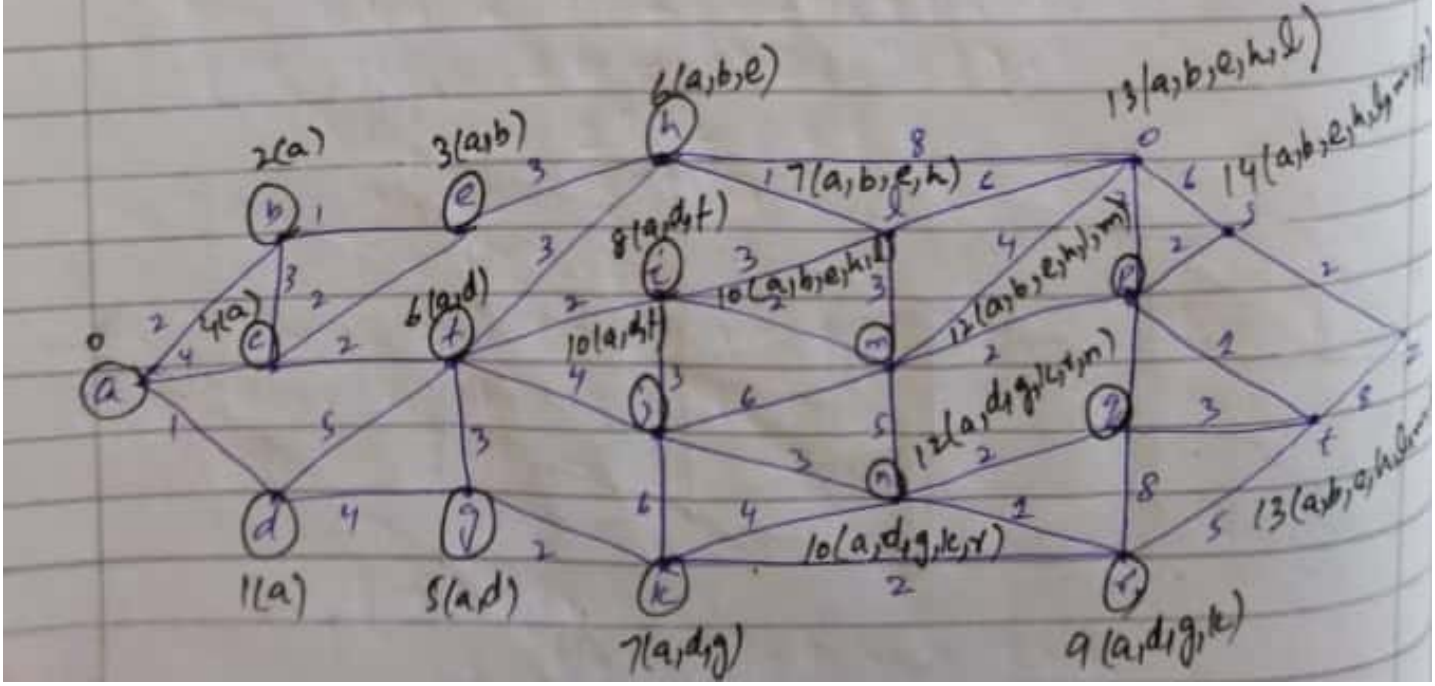




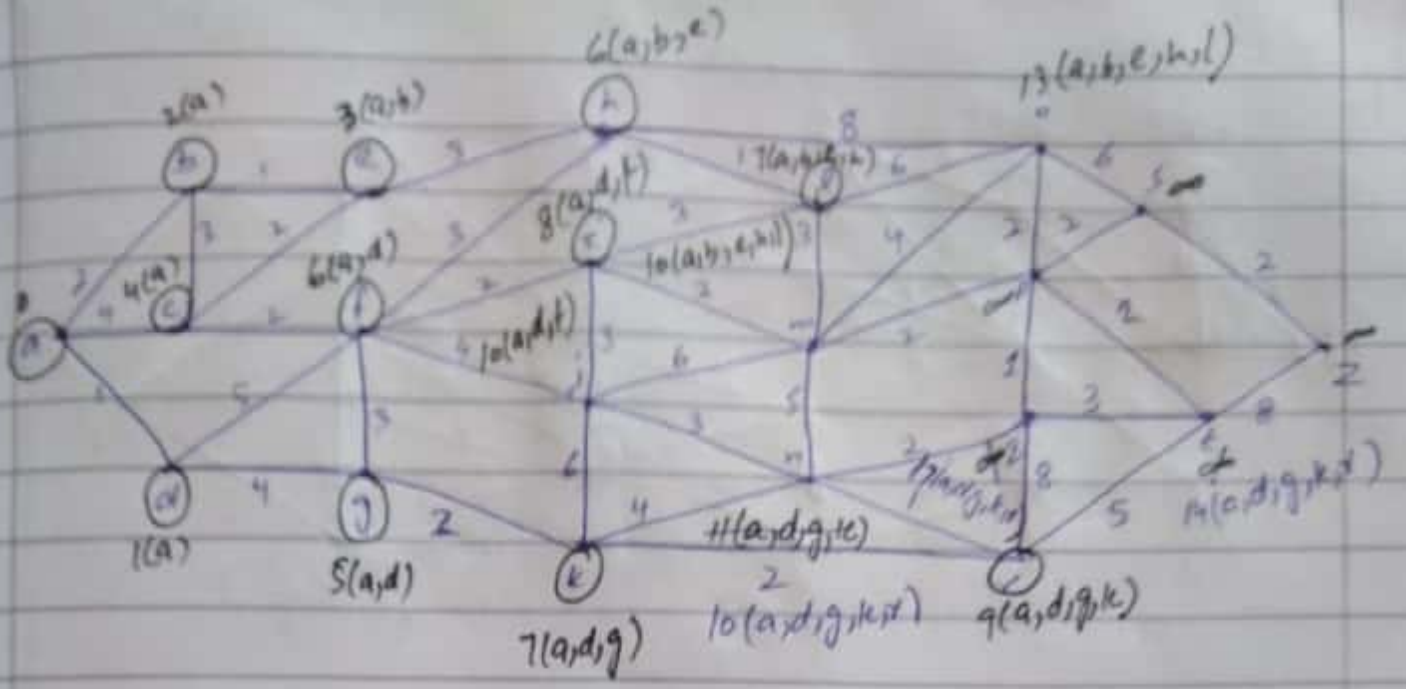
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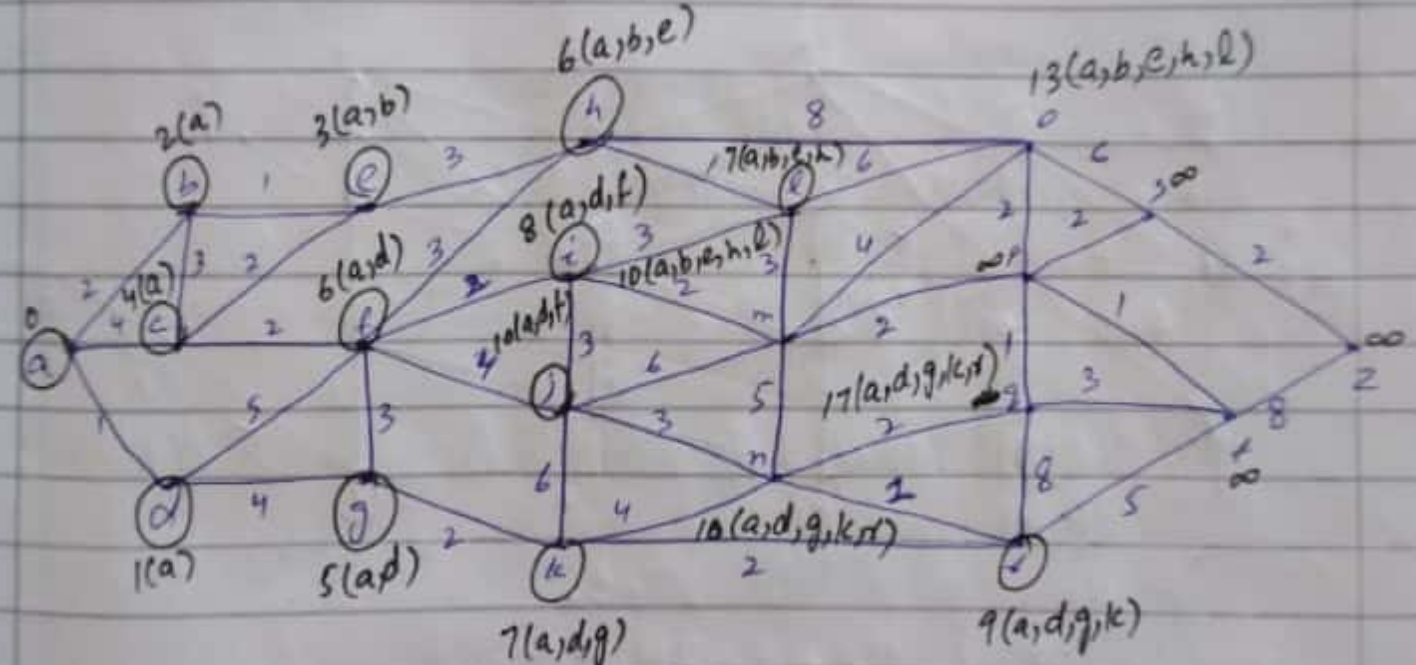


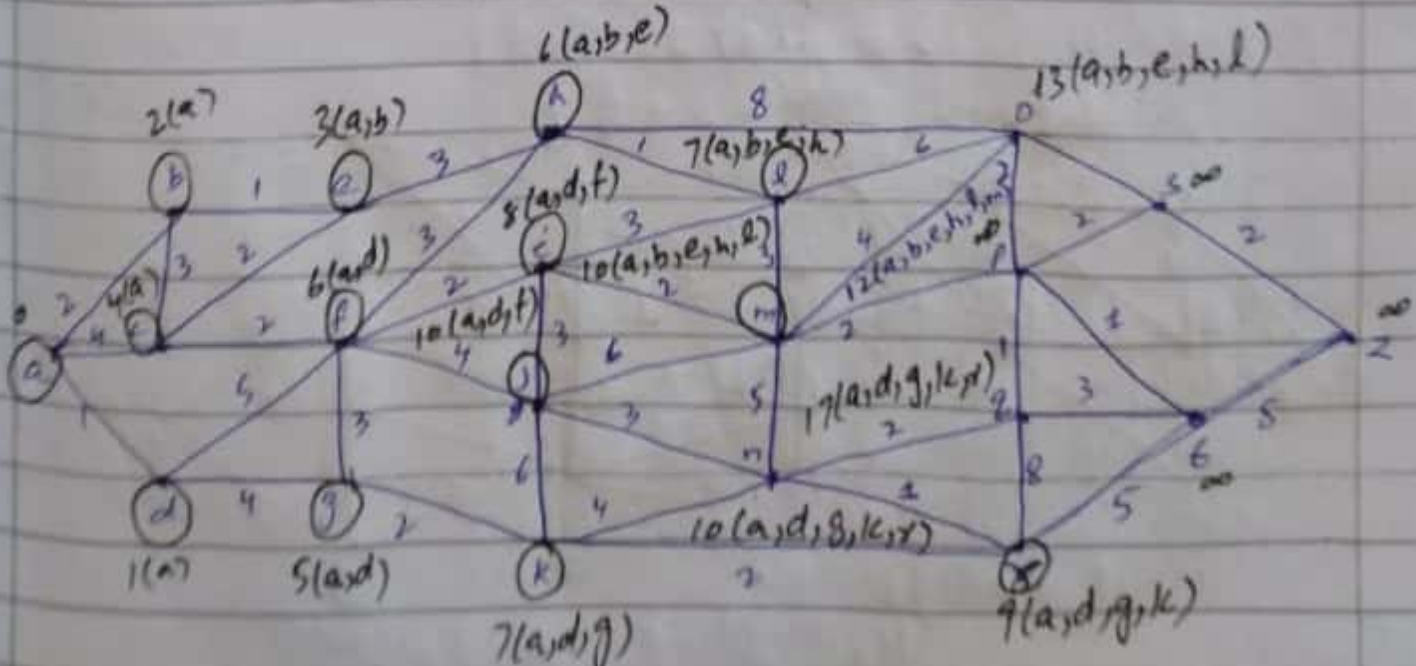




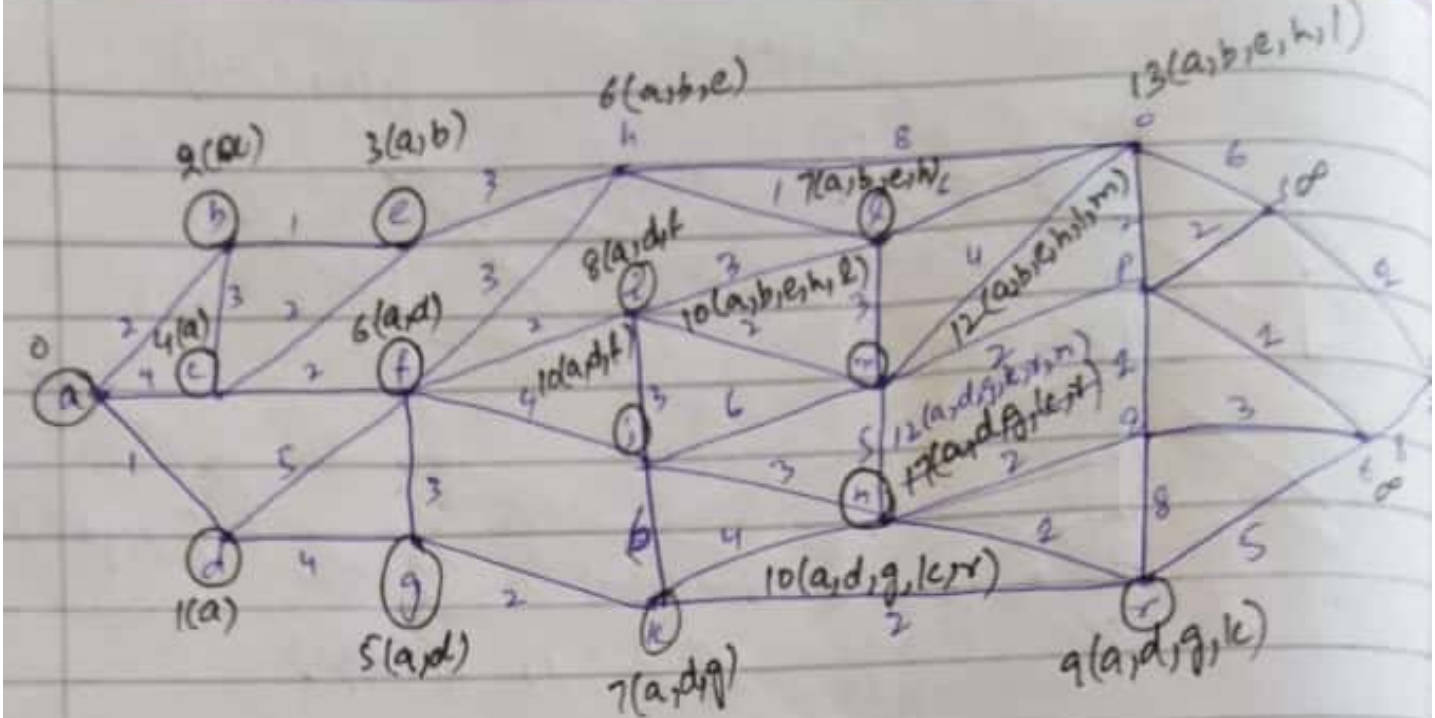
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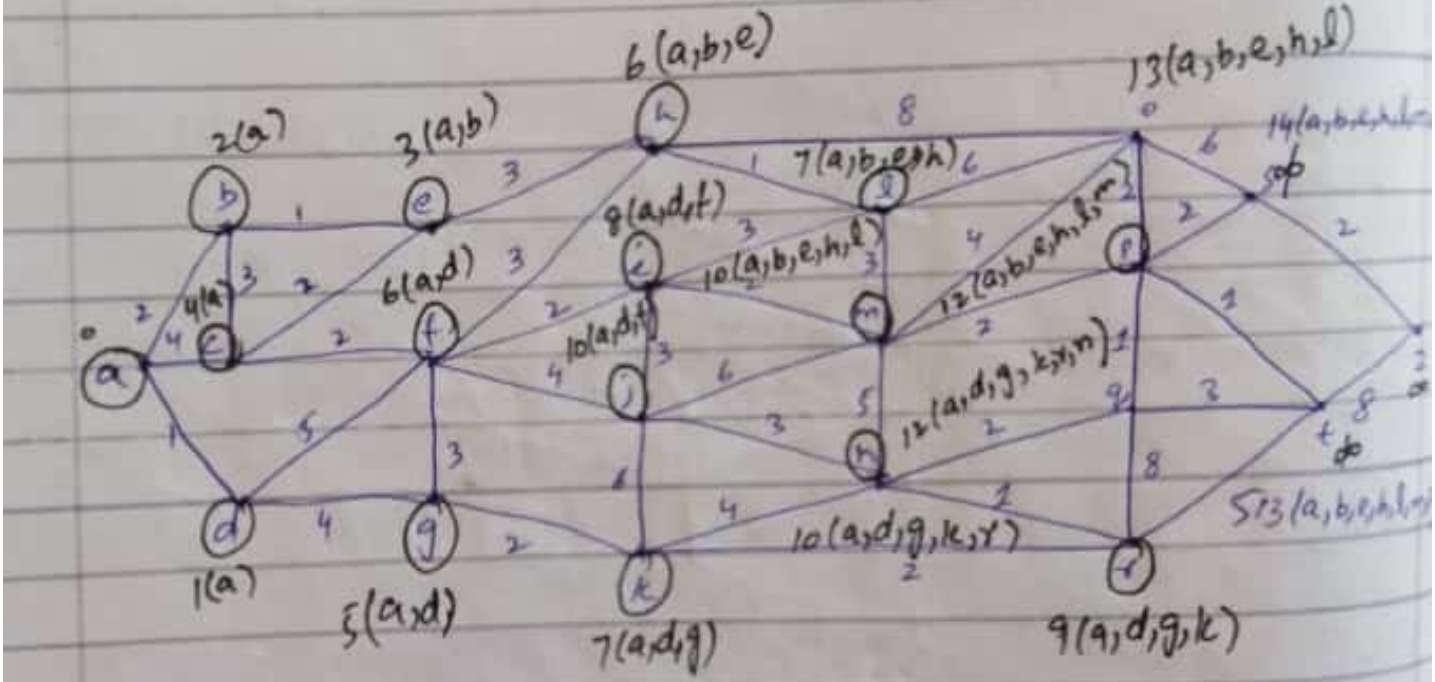


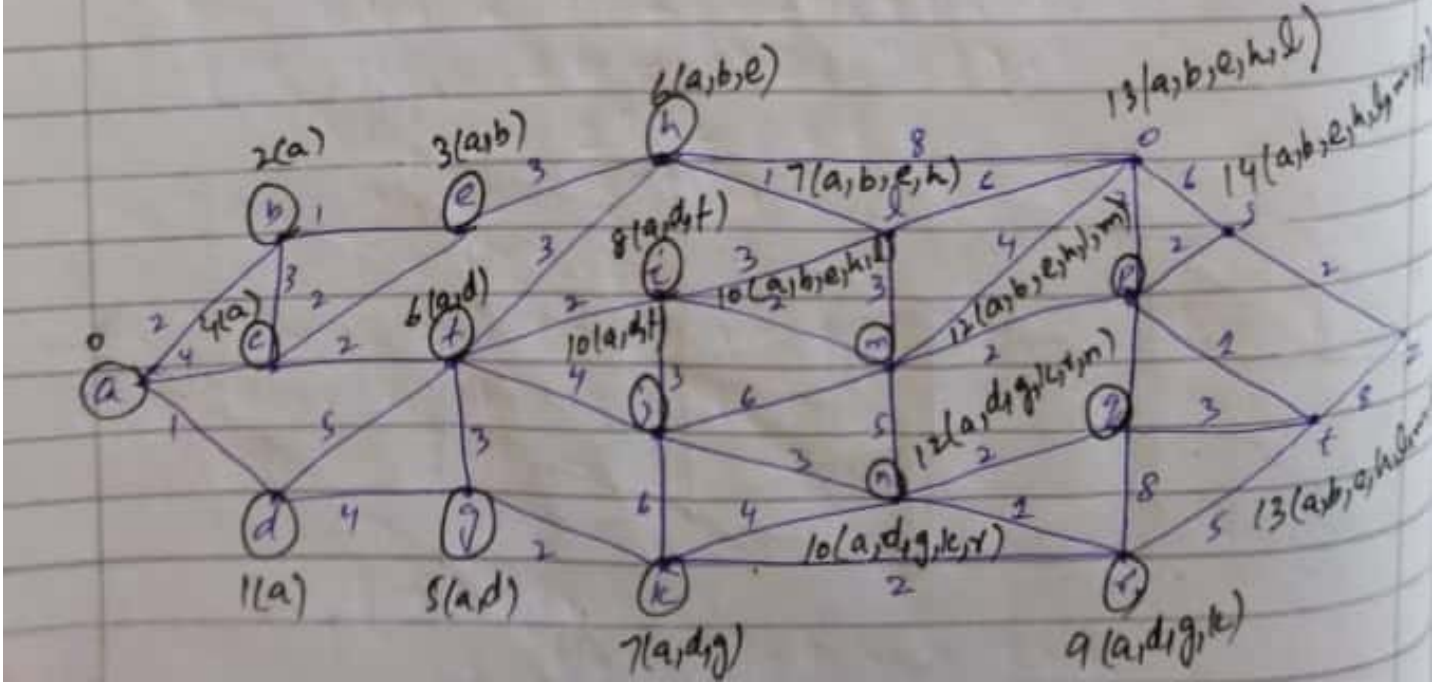




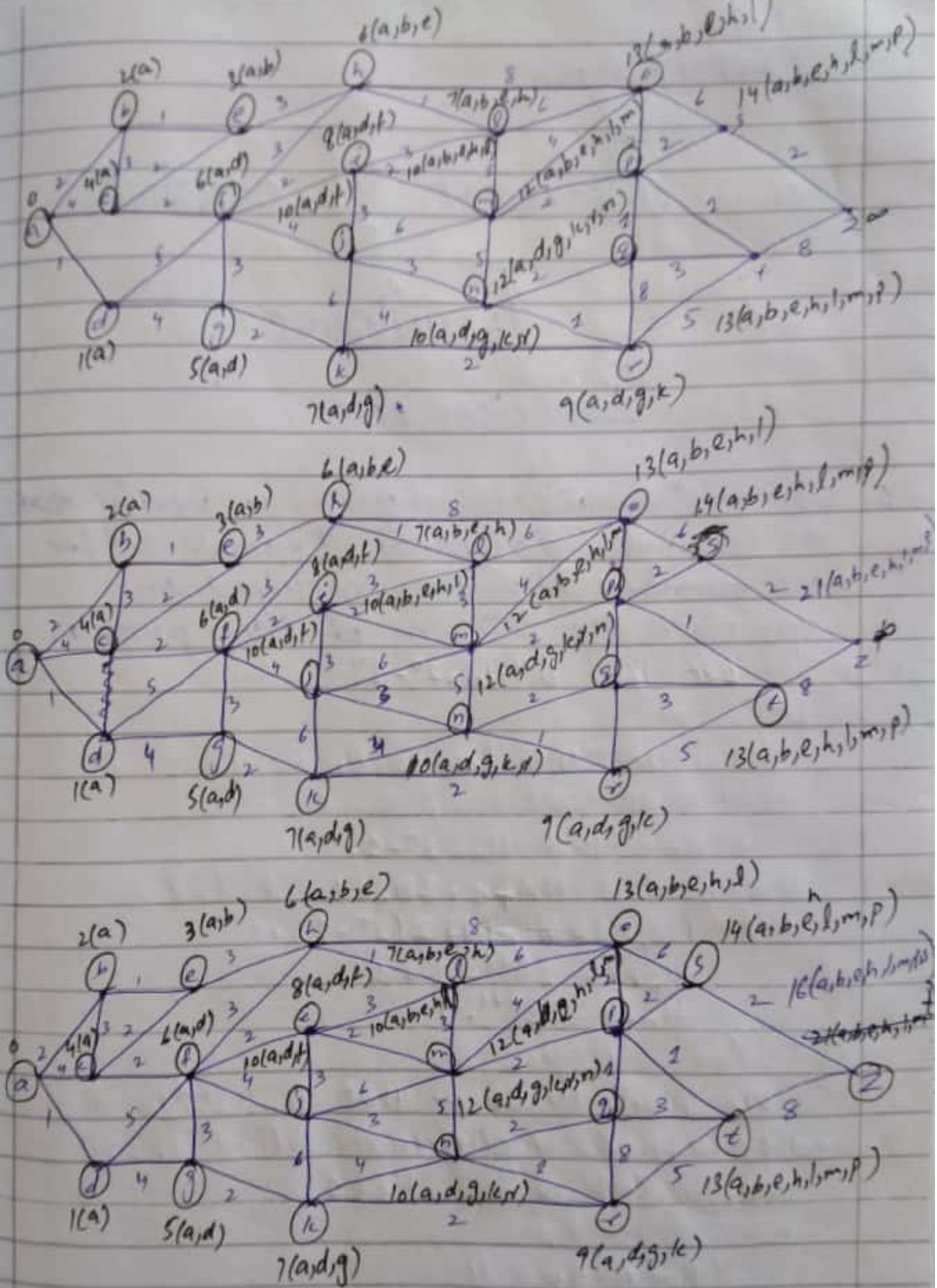
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Scanned with CamScanner



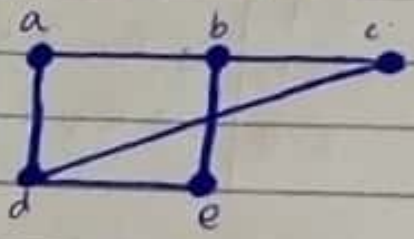
a,b,e,h,l,m,p,s,z is a shortest path whose length is 16.

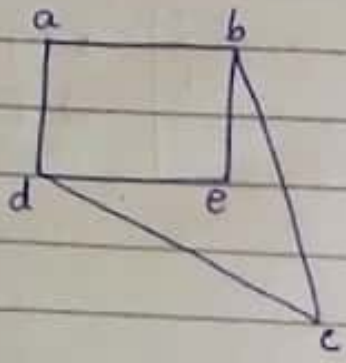
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Exercise 10.7

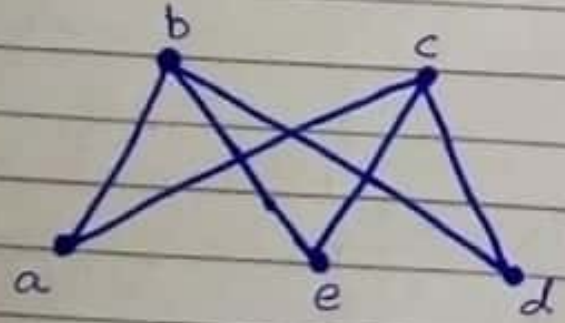
Q: Draw the given planar graph without any crossings.

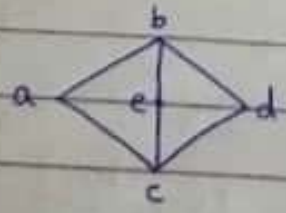
2:



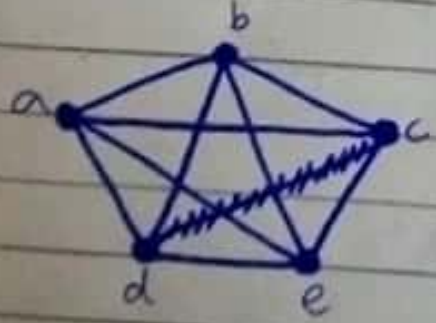


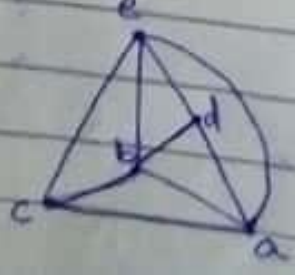
3:





4:

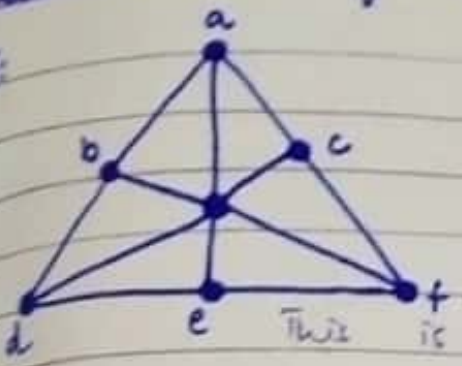




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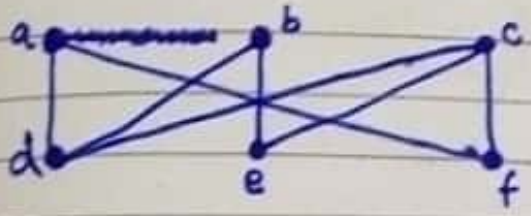
Determine whether given graph is planer if so draw it so that no edge cross.

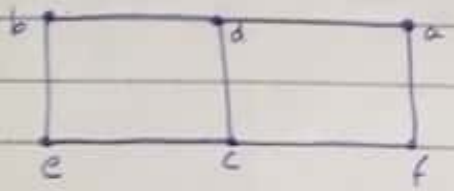
5:



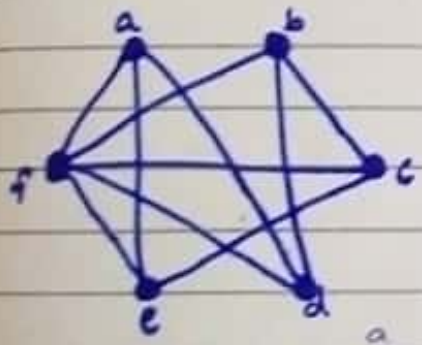
This is not planar graph.

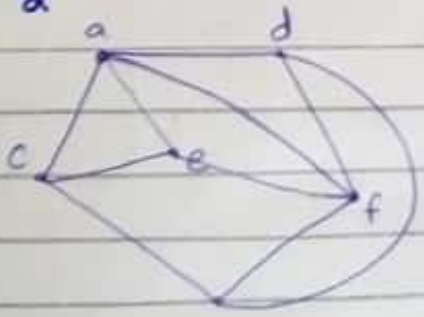
6:



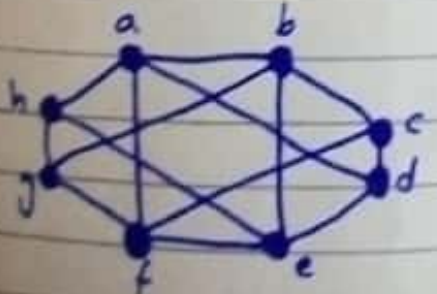


7:



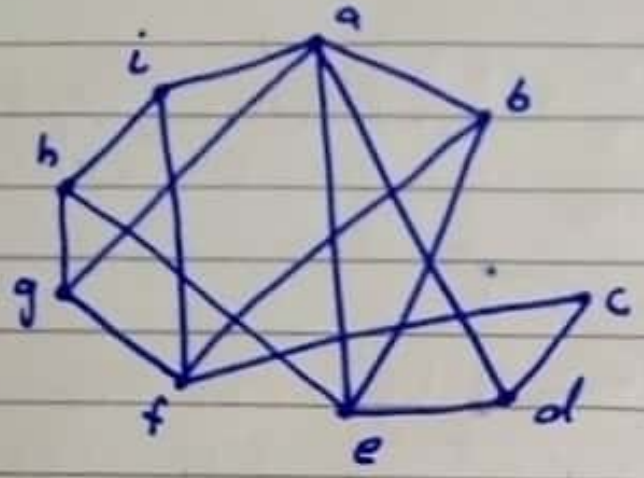


8:



Scanned with CamScanner

9:

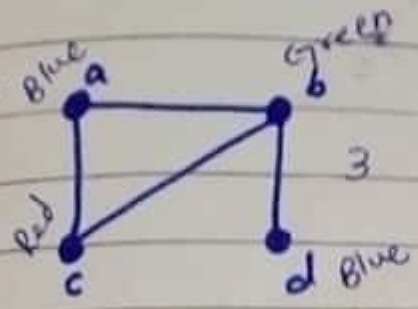


This is not planer graph.

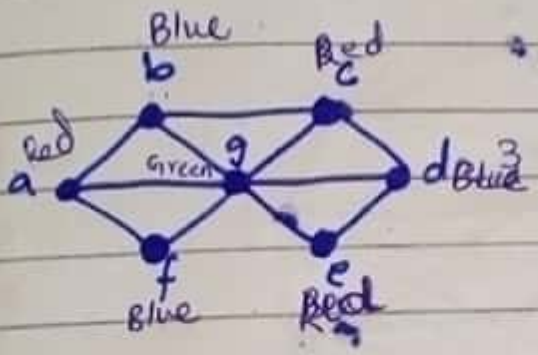
Scanned with CamScanner

Exercise 10.8

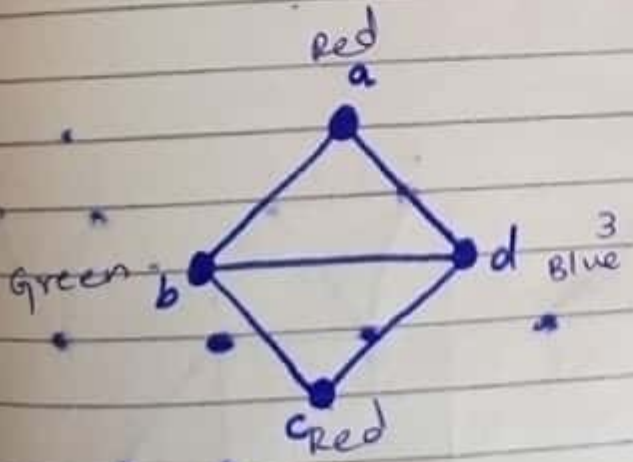
5:



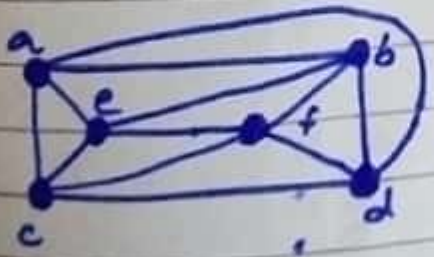
6:



7:



8:

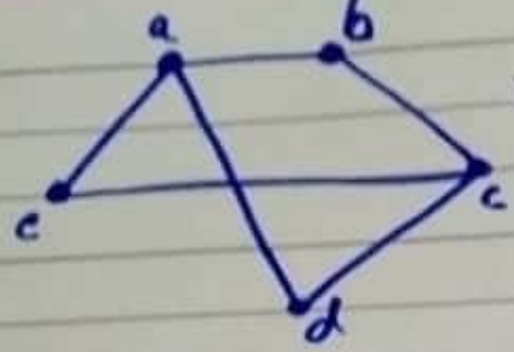


3

a, f = red
e, d = green
c, b, Blue.

Scanned with CamScanner

7:

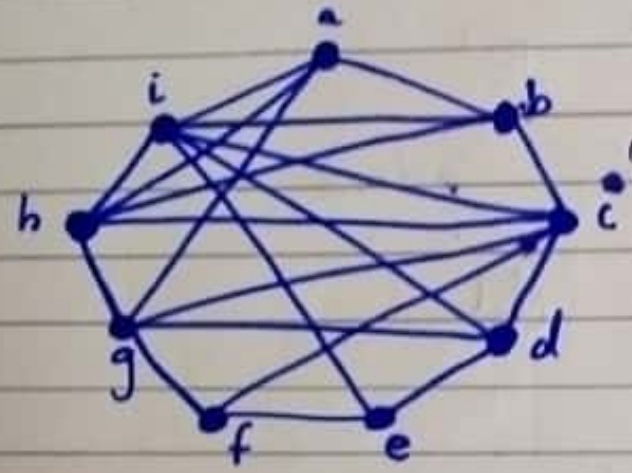


2

b, d, e = red

c, a = blue

10



4

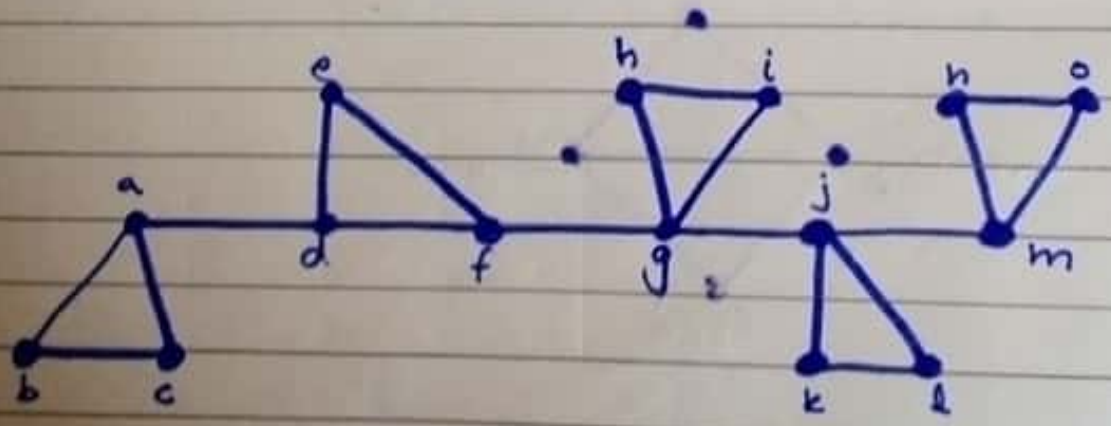
Red = a, c, d, e

Blue = i, g, f

Green = h

Black = b

11:



2

3

Red = a, e, h, n, j

Blue = b, d, i, o, k

Green = c, f, g, l, m

Scanned with CamScanner